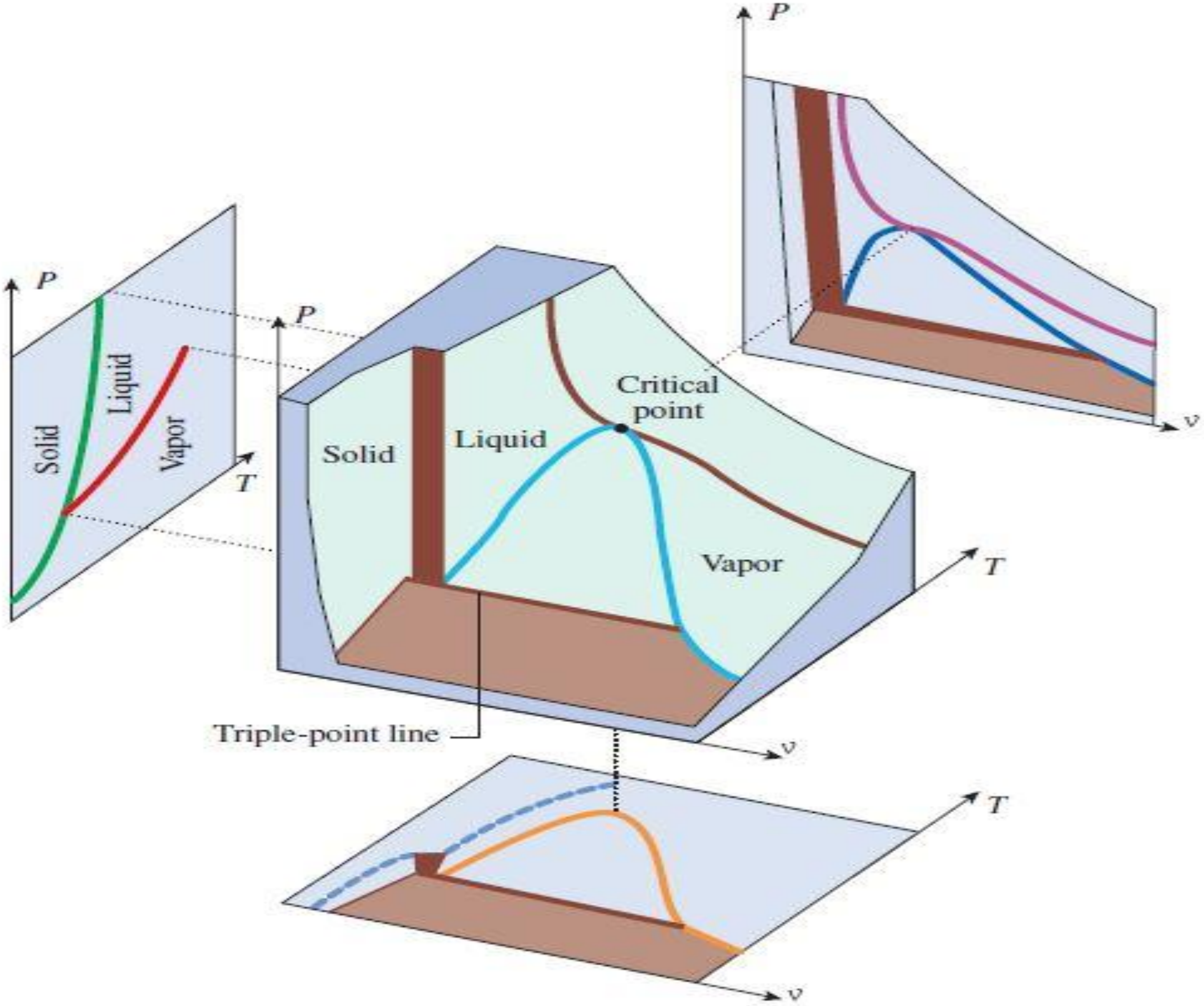


Module-2: Properties of Pure Substance

Thermodynamics (NMEC103) : (Winter 2024-25)

Dr. S. Sahoo, Mechanical Engineering, IIT Dhanbad

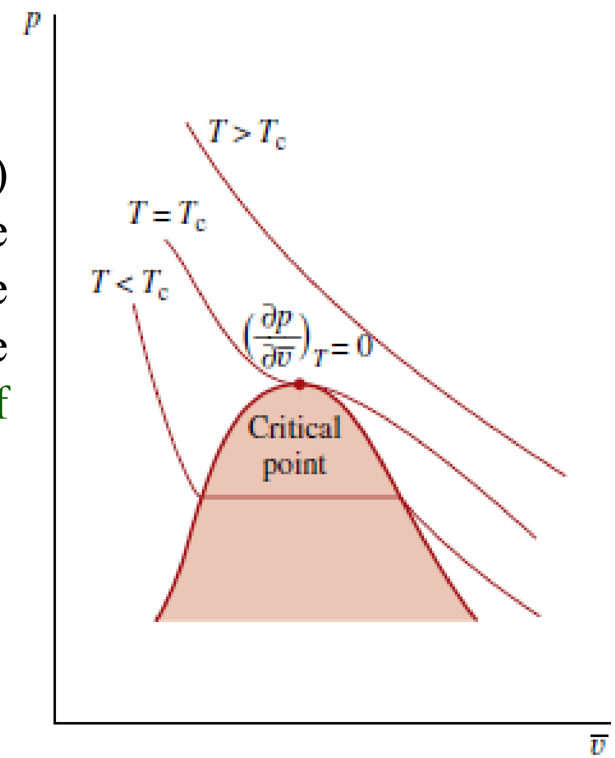
What is so critical about the critical point

- This is a point of discontinuity on the vapor dome. Several thermodynamic properties have discontinuity at the critical point.
- β. Cp and k becomes infinity at the critical point.
- Near the critical point a transparent substance will become almost opaque to light scattering caused by the large fluctuations in the local density This phenomena is called critical opalescence.
- Surface tension decreases with increase in temperature and become zero at the critical point

$$\gamma = \gamma_0 \left(1 - \frac{T}{T_{cr}}\right)^n$$

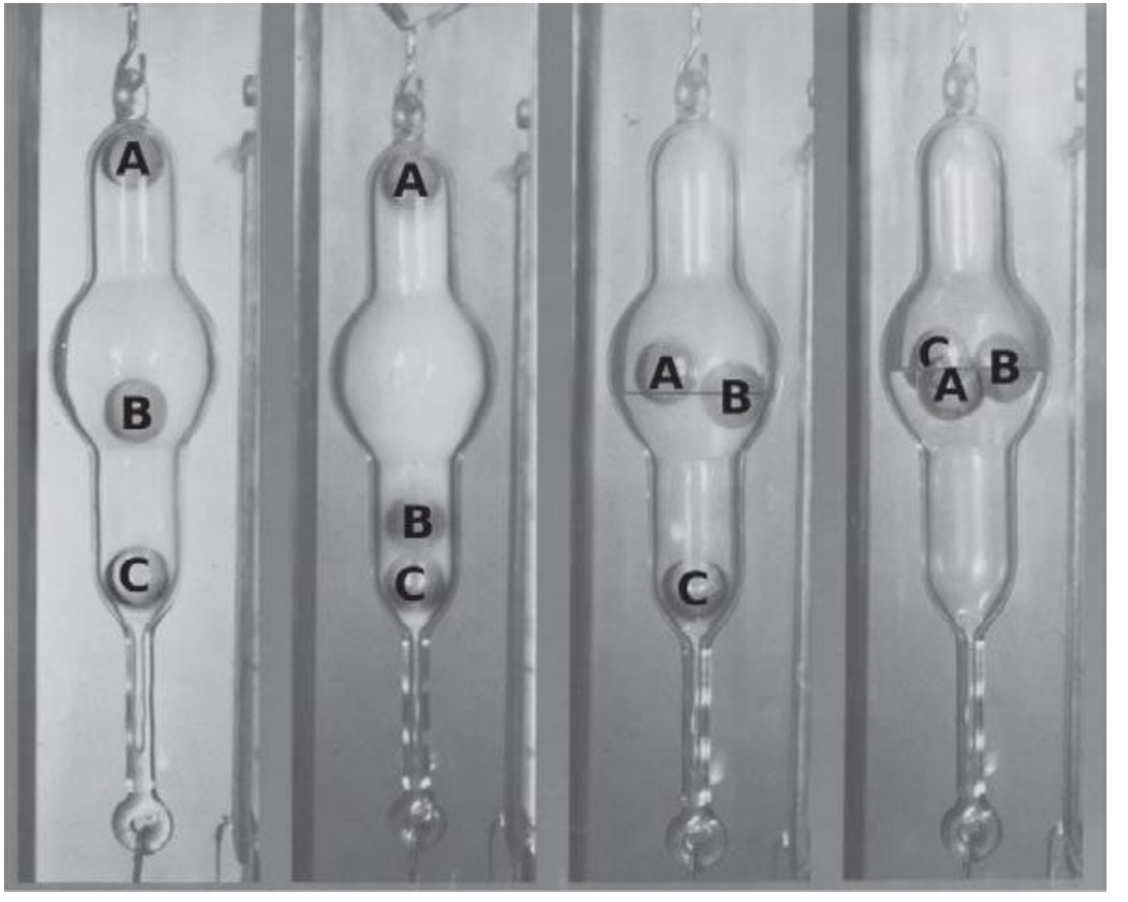
- At the critical point the two volumes (i.e. the v_g and v_f) coincide . On a p-v diagram the critical point is the limiting position as two points lying on a horizontal line approach each other. Hence, at the critical point, the critical isotherm has a horizontal tangent (point of inflection). This can be expressed mathematically as:

$$\left(\frac{\partial p}{\partial \bar{v}}\right)_{T=Tc} = 0; \quad \left(\frac{\partial^2 p}{\partial \bar{v}^2}\right)_{T=Tc} = 0$$



https://www.youtube.com/watch?v=RmaJVxafesU

The glass bulb contains carbon dioxide near the critical density $\rho_{critical}$, and three balls with densities $\rho_A \lesssim \rho_{critical}$ $\rho_B = \rho_{critical}$, and $\rho_C \gtrsim \rho_{critical}$. In (a), the temperature is well above the critical temperature, leaving all the carbon dioxide in the gaseous state. In (b), the temperature is only slightly above the critical temperature and the carbon dioxide has become foggy. In (c), the temperature is slightly below the critical temperature and a meniscus has developed separating the gaseous and liquid states. In (d), the temperature is far below the critical temperature and the density of the liquid has increased to the point where all three balls now float on the surface of the liquid.

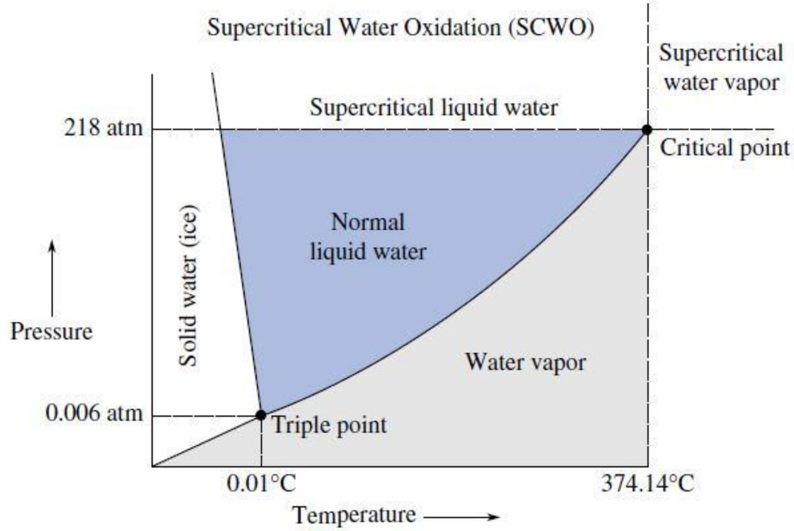


Critical Opalescence

Robert T. Balmer

Waste Disposal Using Supercritical Fluids

- A major engineering challenge today is the development of an effective waste disposal or destruction technique that produces no toxic waste or emission itself. Hazardous wastes have historically been discarded in ocean dumping, landfills, incineration, or long-term storage. However, ocean dumping is now illegal and landfill sites are becoming increasing difficult to manage due to concerns about groundwater contamination.
- A new technology has emerged as an effective way to eliminate thousands of tons of organic wastes that are the by-products of modern society. Called supercritical water oxidation, the technique exploits the fact that, while most organic wastes are not soluble in water at normal temperatures, they are dissolved at high pressure and temperature (Figure 1)



Once the organic wastes are dissolved, oxygen is added and an oxidation-reduction reaction occurs, much like a controlled combustion process. Operating at supercritical conditions results in a single-phase homogenous reaction environment that causes rapid oxidation of the organics, producing carbon dioxide, water, nitrogen, and small amounts of other compounds, such as ammonia and acids. Since the entire process is in a closed system, no harmful products are released into the environment

This technique has also been found to be an effective disposal method for surplus military chemical wastes, such as nerve agents, mustard gas, rocket fuels, TNT, and other explosives. More than 99.9% of the explosives are destroyed in less than 30 s at 600.°C. Even radioactive wastes can be concentrated and stabilized by eliminating their organic components. The resulting radioactive components can then be encased in molten glass and stored deep underground.

Triple point:

- Triple point: any three phases (generally solid, liquid, vapor, but it can be 3 different solid, liquid phases) co-exists.

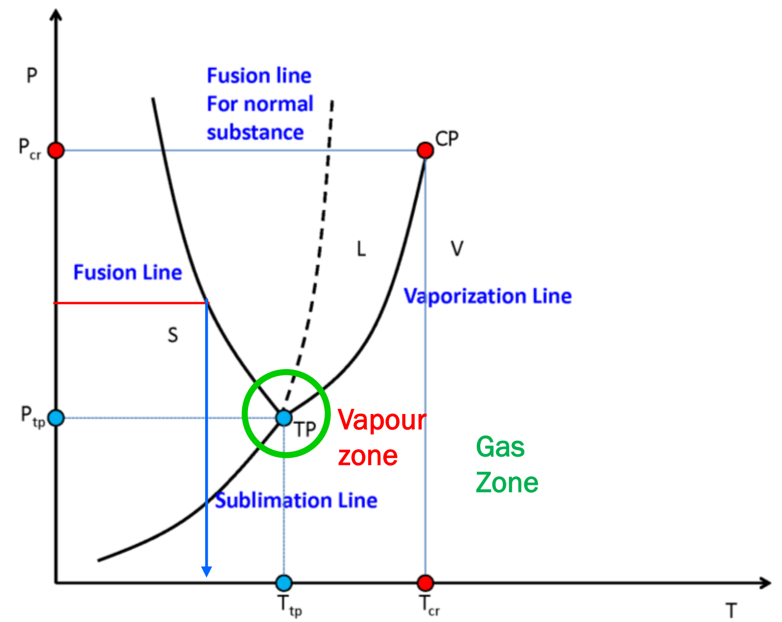
- On the phase diagram, i.e., p-T it is a point (as neither p nor T varies during phase transition) on p-*v* or T-*v* diagram it is a line (**triple phase line**) as "*v*" varies during phase change , on u-v plot it is expanded to an area, the reason being that both u and v changes during phase transition

- Below the triple point the liquid phase does not exist and it is a fixed point and has zero degree of freedom

- It is an easily reproducible state, hence used as a reference point in thermometry

- Consider a cyclic process performed around the triple point and close enough to it so that the only changes in enthalpy occur during phase transitions. Let the substance, initially in the solid phase, be first transformed to the vapor phase, then to the liquid phase, and finally returned to its initial state in the solid phase. There is a heat flow into the system in the first process and the increase in specific enthalpy ,In the second and third processes there is a heat flow out of the system, and the corresponding changes in enthalpy are Then since

$$l_{13} = l_{12} + l_{23}$$



- Enthalpy of Sublimation = Enthalpy of fusion + enthalpy of vaporization at the triple point

Work Done during Phase Change Process

Changes of phase are always associated with changes in volume, so that work is always done on or by a system in a phase change (except at the critical point, where the specific volumes of liquid and vapor are equal). If the change takes place at constant temperature, the pressure is constant also and the specific work done by the system is therefore

Then from the first law, the change in specific internal energy is

$$q_{1-2} = w_{1-2} + (u_2 - u_1) \ldots (1)$$

$$w_{1-2} = p(v_2 - v_1) \ldots \ldots (2)$$

$$q_{1-2} = l \ldots..(3)$$

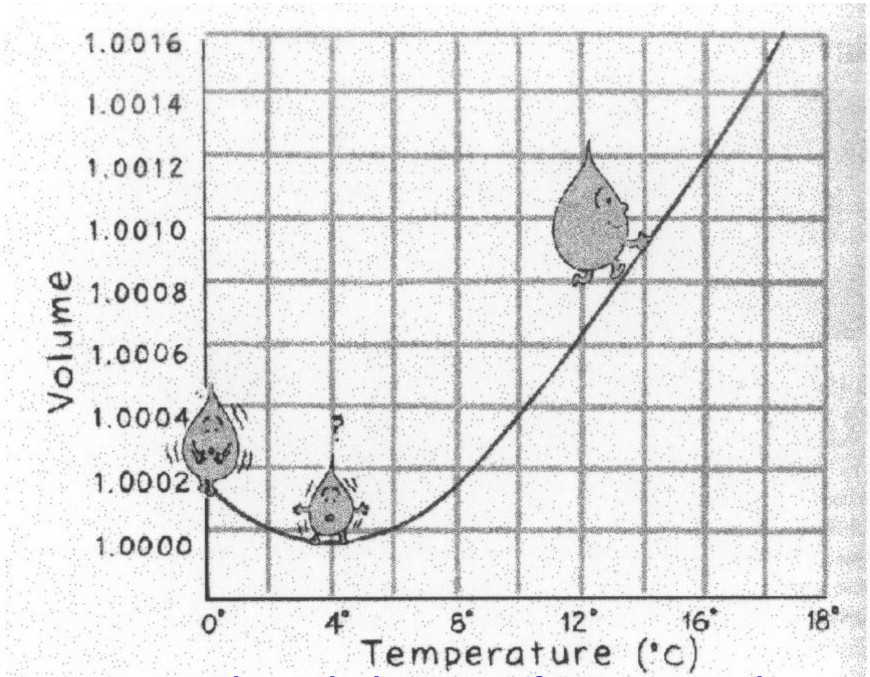
$$u_2 - u_{1=} = l - p(v_2 - v_1) \ldots (4)$$

$$l = (u_2 + pv_2) - (u_1 + pv_1) \ldots.(5)$$

$$l = (u_g + p_{sat} v_g) - (u_f + p_{sat} v_f) \ldots..(6)$$

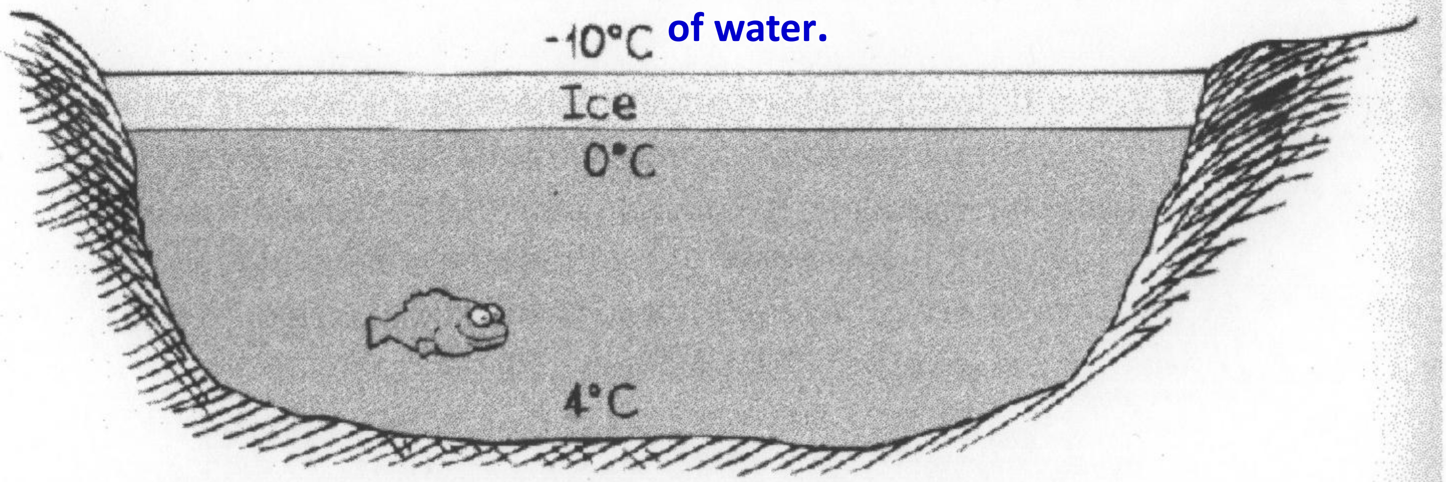
$$l = (h_g - h_f)$$

The Anomalous Behaviour of Water



- ✓ Bursting of water pipes
- ✓ Aquatic life in the polar region
- ✓ Ice Skating

Anomalous behavior of water is the unusual property of water to expand when cooled below 4°C and contract when heated from 0°C to 4°C. This is also known as anomalous expansion of water.

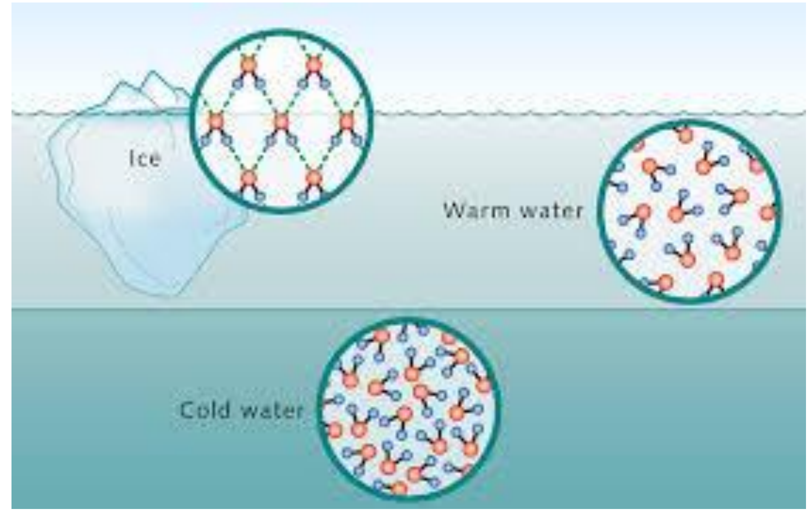


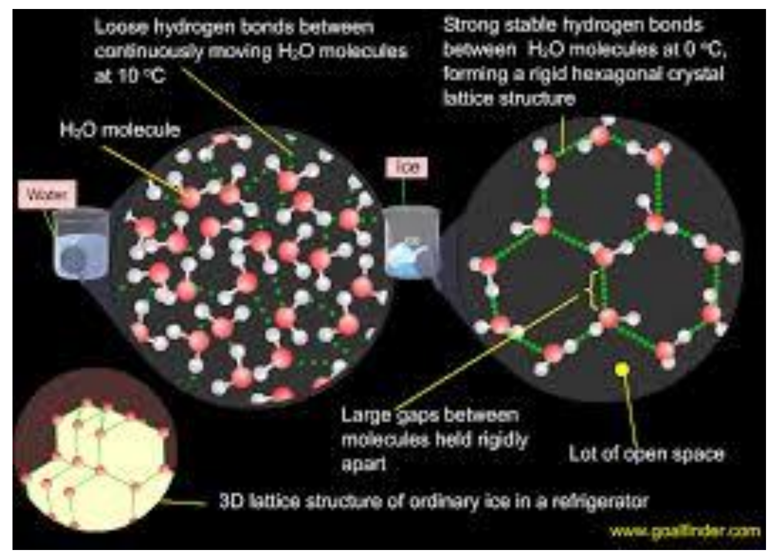
EFFECTS OF ANAMALOUS EXPANSION OF WATER

(b) Icebergs

Since the density of ice ($0.92 g/cm^3$) is slightly less than that of water it floats with only a small portion above the water surface.

The rest and bigger portion rests under water. A bi



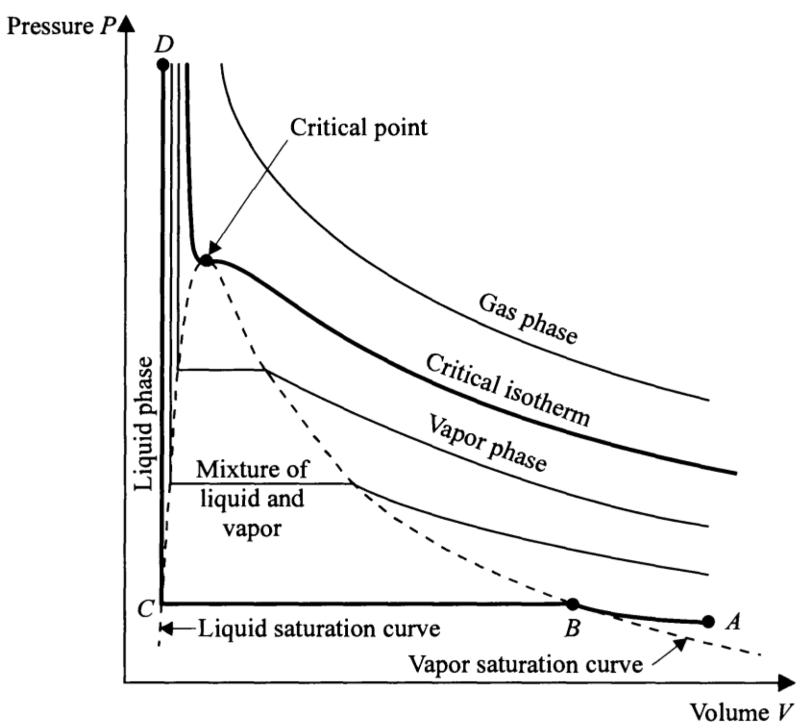


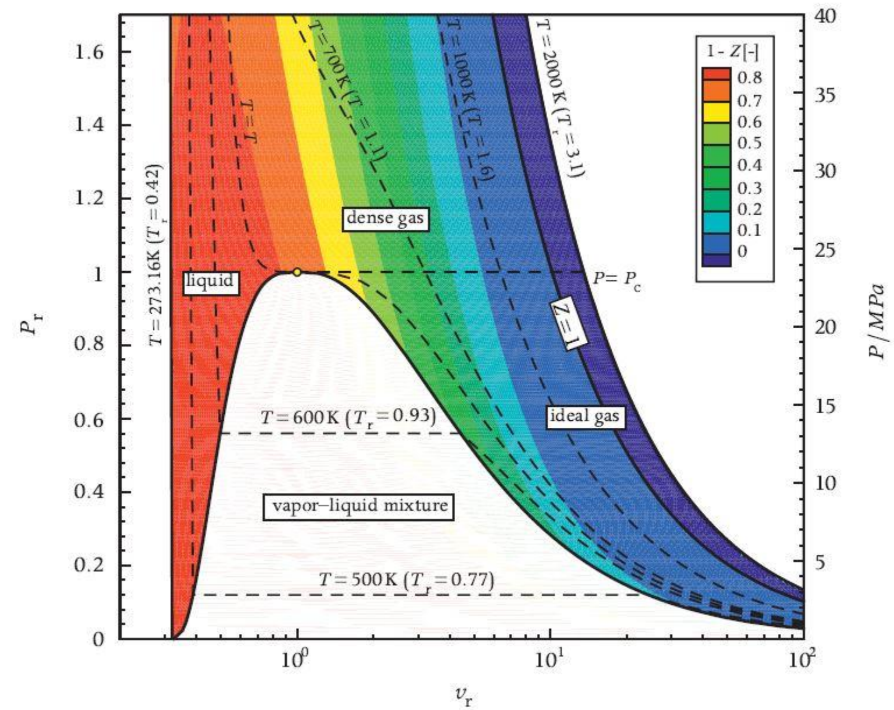
Difference Between Gas & Vapor

The properties of gas and vapor are similar, however with an exception

I. A vapor can be liquified isothermally by increasing the pressure, however a gas can not be liquified by isothermal compression , no matter how high the pressure is.

II. At large volume & low pressures, the isotherms on a $p - v$ plot are equilateral hyperbolas $pv = RT$



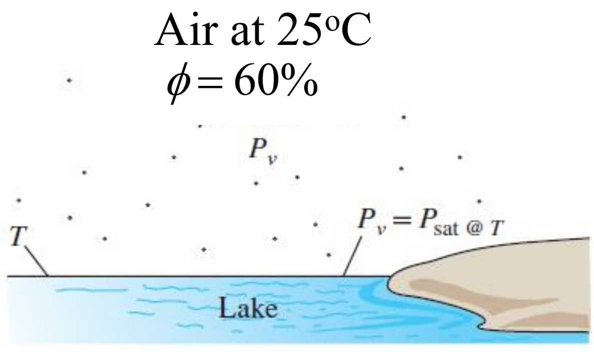


(a) P–v state diagram for water.

Boiling and Evaporation (Both results in vaporization)

Though both boiling & evaporation involve a change of phase from liquid to vapor, there are some differences: Evaporation occurs at the liquid vapor interface when the saturation pressure corresponding to the liquid temperature is greater than the vapor pressure exerted on the liquid.

$$P_{sat}(T) > P_v$$

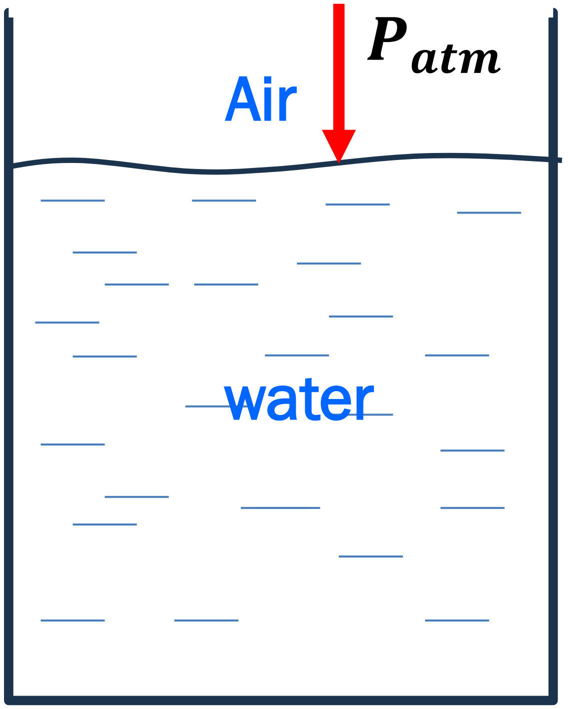


The driving for this process is the difference in saturated & vapor pressures , i.e., $(P\_sat(T) - P_v)$

If the vapor above the liquid is a mixture, then $P_v = partial\ pressure$

$$P_{atm} = P_{O_2} + P_{N_2} + P_{CO_2} + \cdots P_{H2O}$$

Evaporation takes place



$$P_{sat}(T) > P_{H2O}$$

Sweating from human body

$T_f \approx 36^oC$ $P_{sat}(T = 36^oC) = 5.9792\ kPa$

Let the ambient condition are 45ºC & 40% RH

At 45ºC , $P_{sat}(T = 45^oC) = 9.593\ kPa$

$$P_v = P_{sat} \times \frac{RH}{100} = 9.593 \times 0.4 = 3.8371\ kPa$$

$$Since\ P_{sat}(T = \mathbf{36^oC}) = 5.9792\ kPa > P_v\ @\ \mathbf{45^oC} = 3.8372\ kPa$$

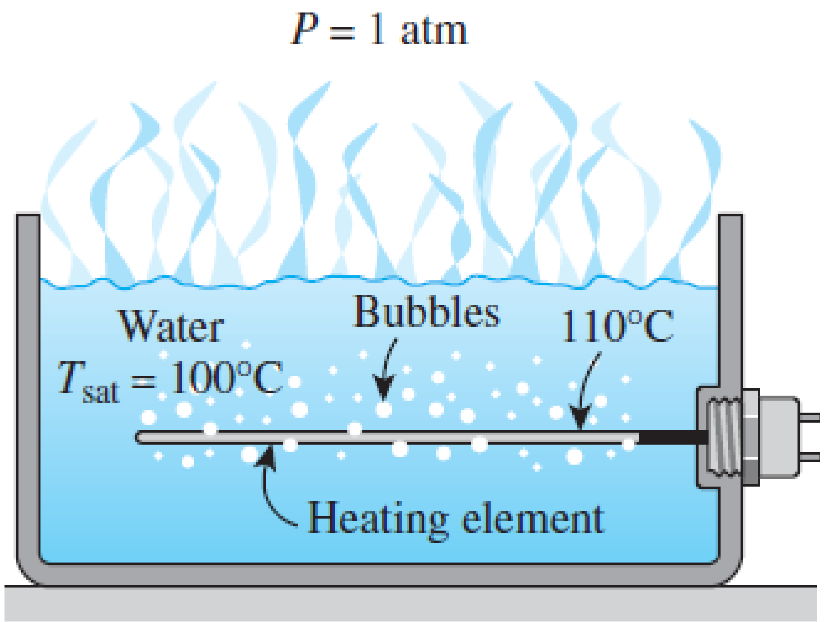
Evaporation of liquid water from human body takes place (body can reject heat even though $T_{amb} > T_{body}$)

At RH=62% $P_v = P_{sat}$@ 36ºC$\Longrightarrow$ No evaporation Fatal

Boiling occurs at solid liquid interface when the temperature of the solid at the interface is much higher than saturation temperature corresponding to the pressure on the liquid surface

$$T_{sur} > T_{sat}(P)$$

At 1 atm, T_{sat} of water =100ºC, so if we have to boil water in a container then the container temperature should be >100ºC



In typical boiling process we see the formation of bubble at the solid liquid interface and then motion due to buoyancy

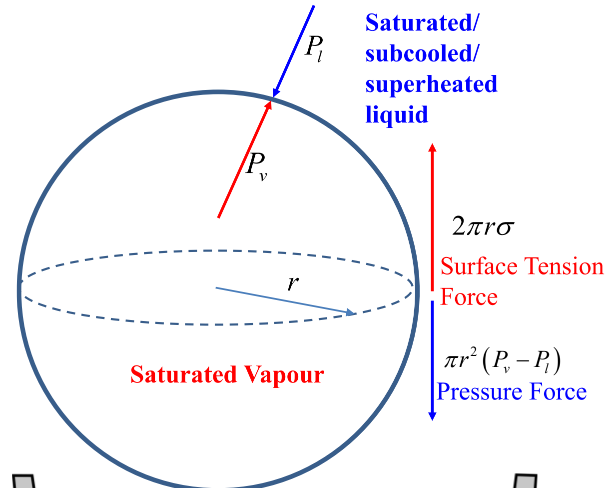
$$P_{sat}(T) > P_{liquid\ surface}$$

Hydrodynamics of the Bubble Formation and Growth in Boiling

- The heat transfer rate during nucleate boiling is influenced by both the agitation caused by the bubbles and the vapour transport of energy.
- Experiments have shown that the bubbles are not always in thermodynamic equilibrium with the surrounding liquids, i.e., the vapour inside the bubble is not necessarily at same temperature as liquid.

From force balance:

$$\pi r^2\left(P_v - P_l\right) = 2\pi r\sigma \Rightarrow \left(P_v - P_l\right) = \frac{2\sigma}{r}$$

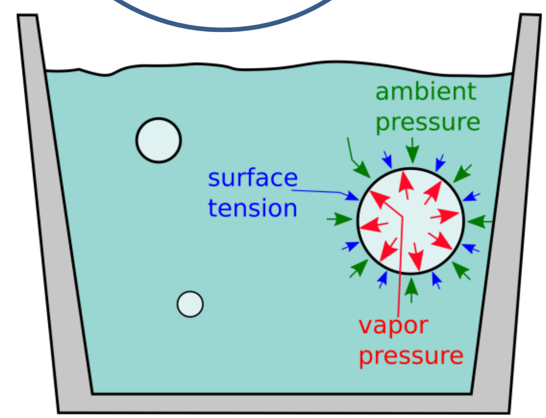


- If the bubble is in pressure equilibrium (i.e., it is neither growing nor collapsing,)

$$r = const \Rightarrow \left(P_v - P_l\right) = const$$

- If the temperature of the vapour inside the bubble is corresponding to the saturation temperature at P_v and the temperature of the liquid is corresponding to the saturation temperature at pressure P_l.

$$T_{v,inside} > T_{l,liquid}\left(\because P_v > P_l\right)$$

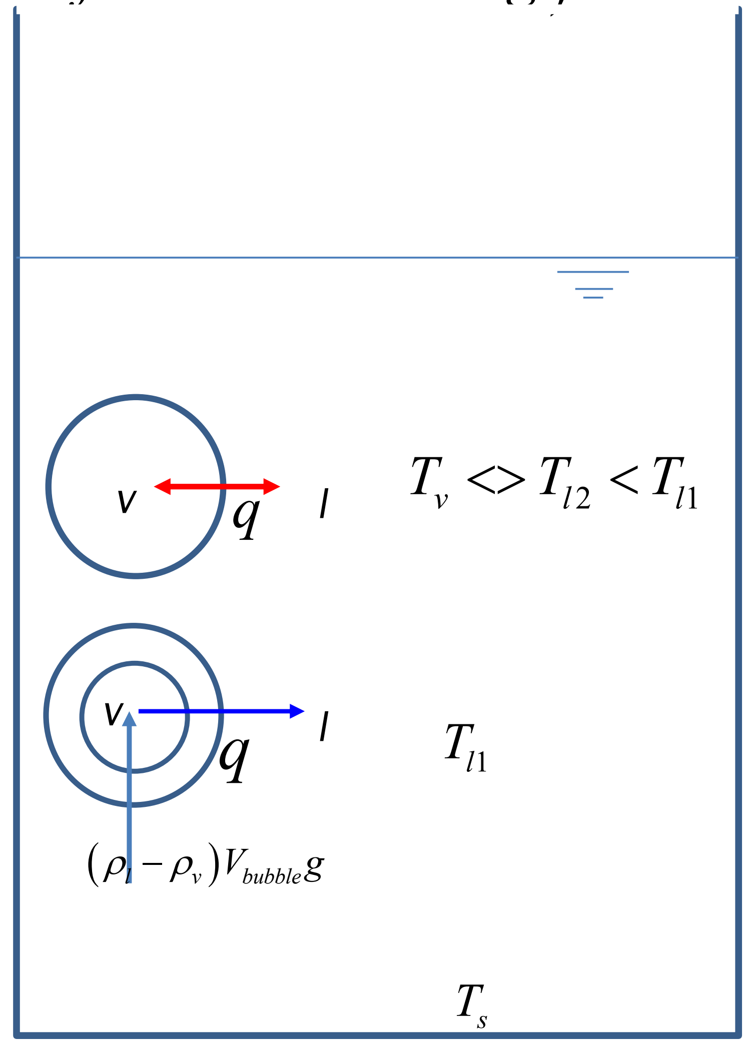


Heat is conducted out of the bubble, the vapour inside condenses and the bubble must collapse. This is what happens when the bubble collapse at the heating surface or in the pool of liquid

- For bubble to grow and escape the surface they must receive heat from the liquid i.e., the liquid must be superheated (**Metastable state**) . So that

$$T_{v,inside} < T_{l,liquid}\left(Observed\ in\ some\ region\ of\ Nucleate\ boiling\right)$$

- As the heat is conducted to the bubble from liquid, evaporation of liquid takes place at the liquid vapour interface (increases the vapor volume).Hence r increases resulting a decrease in the excess pressure
- If P_l is constant P_v must reduce, hence T_v must reduce and heat transfer between the liquid and vapour increases if the vapour stays in the same place. However, as the volume of the bubble increases due to increase in the buoyancy, the bubble moves up into the regions of lower liquid temperature. *Depending upon the superheated liquid temperature the bubble may rise to the surface or may collapse as it moves into colder liquid region.*

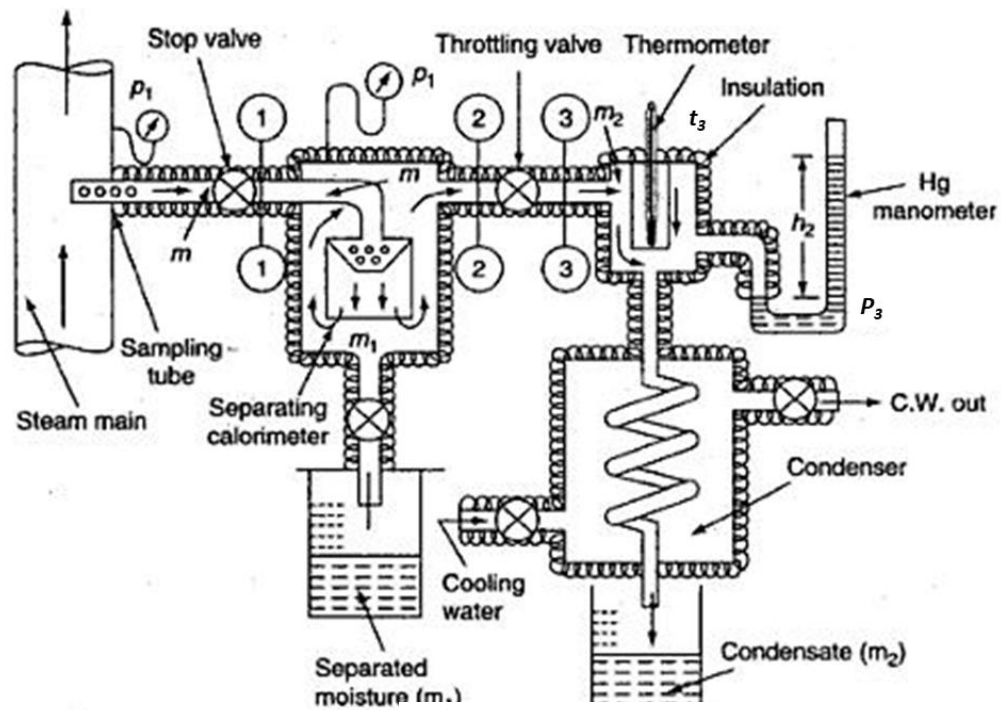


$$\because \pi r^2\left(P_v - P_l\right) = 2\pi r\sigma \Rightarrow \left(P_v - P_l\right) = \frac{2\sigma}{r}$$

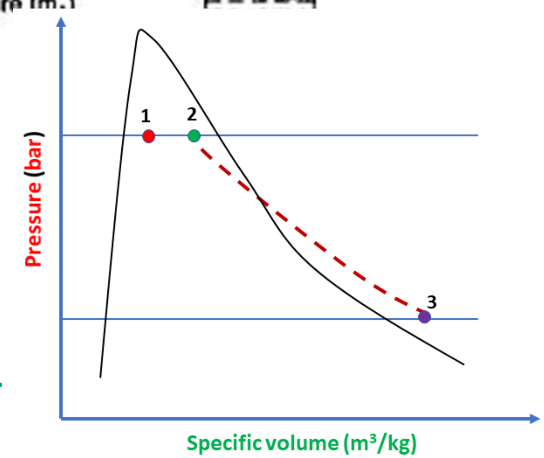
Determination of Dryness Fraction of liquid vapor mixture

Dryness fraction is defined as the ratio of the mass of the saturated vapor present to the total mass of the steam (mixture of saturated liquid & vapor) that contains it

$$x = \frac{m_s}{m_s + m_w}$$



From figure, it can be seen that process $1-2$ represents the moisture separation from the wet sample of steam at constant pressure $p_1 = p_2$, and process $2-3$ represents throttling to pressure p_3. With p_3 and T_3 being measured, h_3 can be found from the superheated steam table.



$$h_3 = h_2 = h_{f@p_2} + x_2 h_{fg@p_2} \qquad x_2 = \frac{h_3 - h_{f@p_2}}{h_{fg@p_2}}$$

If m kg of steam is taken through the sampling tube *in*, t secs, m_1 kg of water is separated mechanically, not by condensation, and m_2 kg is throttled and then condensed to water and collected, then

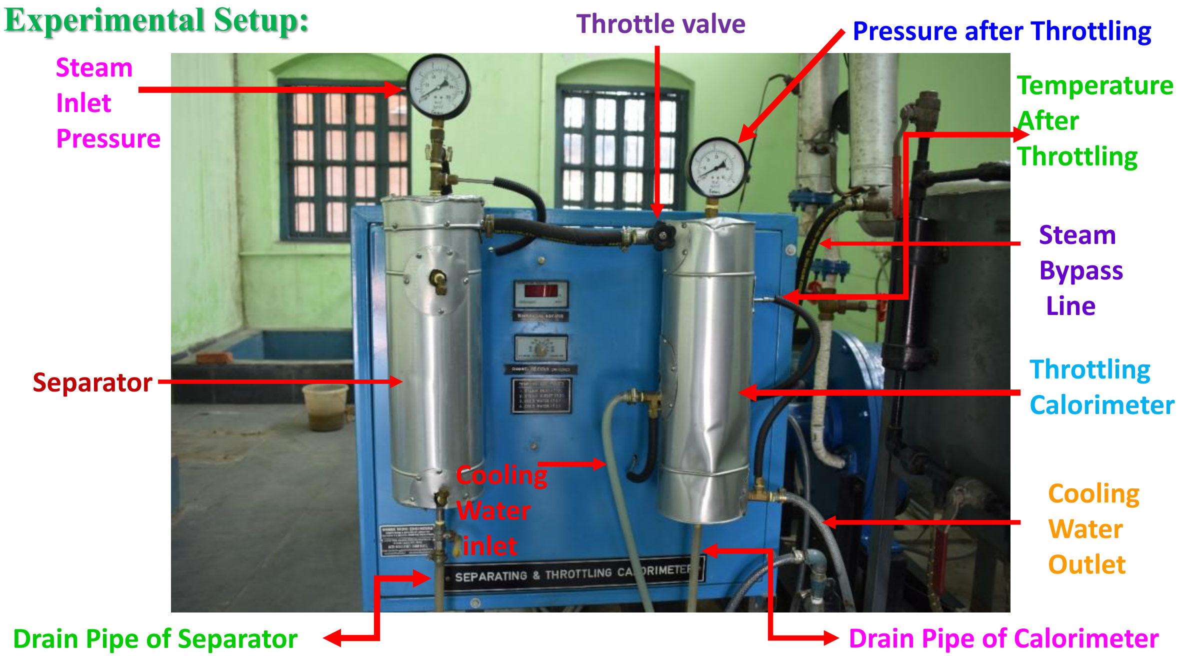
$$\boldsymbol{m = m_1 + m_2,}$$

At state 2, the mass of dry vapor will be $x_2 \times m_2$. Therefore, the quality of the sample of steam at state 1, is given by

$$x_1 = \frac{\textit{mass of dry vapour at state } 2 = \textit{mass of dry vapour at state } 1}{\textit{mass of liquid separated} + \textit{liquid and vapour mixture at state } 2}$$

$$= \frac{x_2 \times m_2}{m_1 + m_2}$$

Experimental Setup:



Evaluation of thermodynamic properties

- Property is any observable or identifiable macroscopic characteristics of a system, It may depend on the extent , size and mass of the system (Extensive) or may be independent (intensive) of the same.
- Properties related to energy interaction $\left(u, h, s, p, T, v, a, g, c_p, c_v, \beta, \kappa, \mu_{JT}\ etc.\right)$ are called the thermodynamic properties.
- We have already seen that thermodynamic properties are related to each other. The state principle, introduced earlier, determines the number of independent properties of a system in a state of stable equilibrium. It also assures that any property can be expressed in terms of the independent properties.
- It may be measurable (pressure, temperature, volume, specific heat, and the Joule –Thomson coefficient are relatively easy to measure) by a measuring device or calculated using the thermodynamic property relations, from the measured properties with the help of the equation of state

- For example, to determine the entropy change between two given states, as defined by we must find a reversible process connecting the two end states and measure the temperatures and differential heat interactions along that path. This is quite a formidable task. The thermodynamic relations that we develop in this chapter allow us to determine entropy and other derived properties from primitive properties, which are easier to measure.

- Properties are the identity of any state of the system.

- Properties are point functions , and their change depends only on the initial and final equilibrium states, no matter what is the path followed by the system to arrive at this final state.

- When we are measuring any measurable property using any gauge it is the average effect of all the molecules constituting the system. This concept of single average value representing the whole system is only valid, when the continuum hypothesis is valid and the system is in thermodynamic equilibrium.

- The deferential of the property is an exact differential which implies that total change in any property is the sum of the contributions from individual changes.

Introduction: A little Math

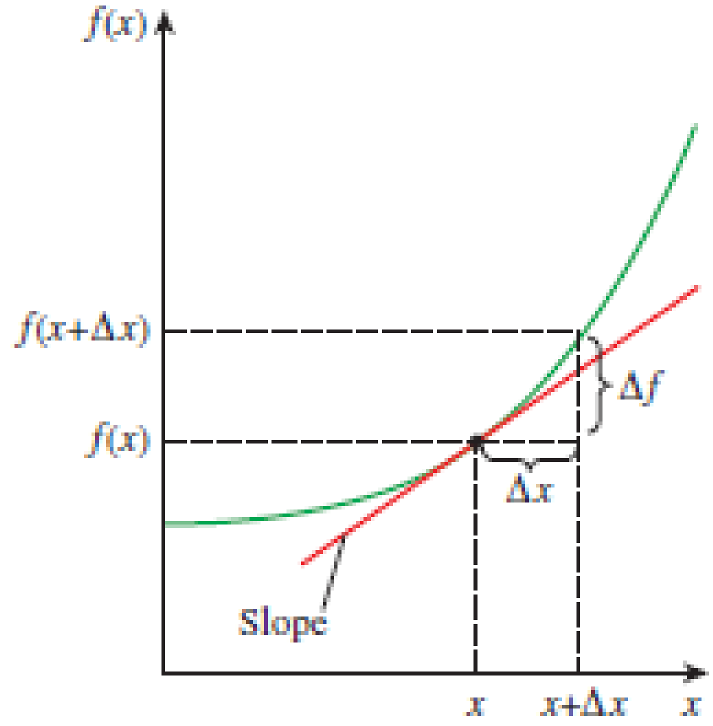
Many of the expressions developed in this chapter are based on the state postulate, which expresses that the state of a simple, compressible substance is completely specified by any two independent, intensive properties. All other properties at that state can be expressed in terms of those two properties. Mathematically speaking,

$$z = f(x, y) \ldots\ldots (1)$$

where x and y are the two independent properties that fix the state and z represents any other property. Most basic thermodynamic relations involve differentials. Therefore, we start by reviewing the derivatives and various relations among derivatives to the extent necessary in this chapter. Consider a function f that depends on a single variable x, i.e.,

$$f = f(x) \ldots\ldots (2)$$

Figure shows such a function that starts out flat but gets rather steep as x increases. The steepness of the curve is a measure of the degree of dependence of f on x. In our case, the function f depends on x more strongly at larger x values. The steepness of a curve at a point is measured by the slope of a line tangent to the curve at that point, and it is equivalent to the **derivative** of the function at that point defined as



$$\frac{df}{dx} = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \quad (First\ Principle(exact))$$

Therefore, *the derivative of a function f(x) with respect to x represents the rate of change of f with x*.

$$\frac{df}{dx} \approx \frac{f(x+\Delta x) - f(x)}{\Delta x} \quad (Finite\ Difference\ (approx))$$

Approximating Differential Quantities by Differences

The C_p of ideal gases depends on temperature only, and it is expressed as.

$$C_p = \frac{dh(T)}{dT}$$

Determine the C_p of air at 300 K, using the enthalpy data given below

	T=305 K	T=295 K
Enthalpy (h)	305.22 kJ/kg	295.17 kJ

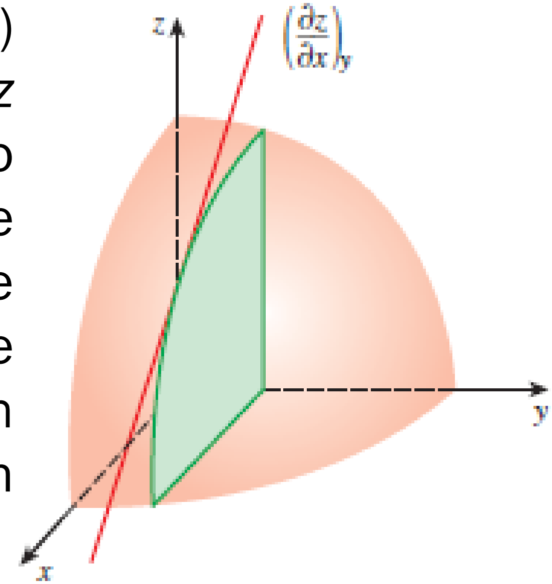
$$C_p(300K) = \left[\frac{dh(T)}{dT}\right]_{300K} \cong \left[\frac{\Delta h(300T)}{\Delta T}\right]_{T=300K} = \frac{h(305K)-h(295K)}{305-295}$$

$$\mathrm{C_p(300K)} \cong \frac{\mathrm{h(305K)-h(295K)}}{305-295} = \frac{305.22-295.17}{305-295} = 1.005\,\frac{\mathrm{kJ}}{\mathrm{kg-K}}$$

Note that the calculated C_p value is identical to the listed value. Therefore, differential quantities can be viewed as differences. They can even be replaced by differences, whenever necessary, to obtain approximate results. The widely used finite difference numerical method is based on this simple principle.

A little Math. Partial Derivative and

Now consider a function that depends on two (or more) variables, such as $z = f(x, y)$. This time the value of *z* depends on both *x* and *y*. It is sometimes desirable to examine the dependence of *z* on only one of the variables. This is done by allowing one variable to change while holding the others constant and observing the change in the function. The variation of *z*(*x*, *y*) with *x* when *y* is held constant is called the **partial derivative** of *z* with respect to *x*, and it is expressed as



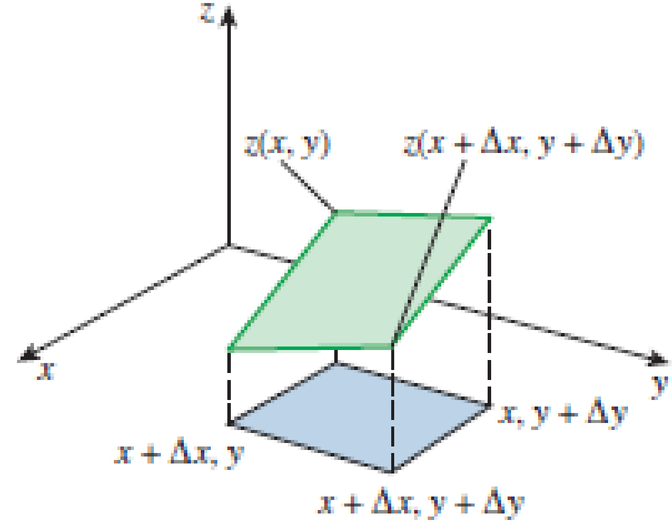
$$\left(\frac{\partial z}{\partial x}\right)_y = \lim_{\Delta x \to 0}\left(\frac{\Delta y}{\Delta x}\right)_y = \lim_{\Delta x \to 0}\frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$$

Note that the changes indicated by d and ∂ are identical for independent variables, but not for dependent variables. For example, $(\partial x\,)_y = dx$; $(\partial z\,)_y \neq dz$. In our case

$$dz = (\partial z\,)_x + (\partial z\,)_y$$

Also note that the value of the partial derivative $\left(\frac{\partial z}{\partial x}\right)_y$ is different at different y value

To obtain a relation for the total differential change in z(x, y) for simultaneous changes in *x* and *y*, consider a small portion of the surface z(x, y) shown in Figure. When the independent variables *x* and *y* change by Δx and Δy respectively, the dependent variable *z* changes by Δz , which can be expressed as



$$\Delta z = z(x + \Delta x, y + \Delta y) - z(x, y)$$

Adding and subtracting $z(x, y + \Delta y)$ we get

$$\Delta z = z(x + \Delta x, y + \Delta y) - z(x, y + \Delta y) + z(x, y + \Delta y) - z(x, y)$$

$$\Delta z = \frac{z(x+\Delta x, y+\Delta y) - z(x, y+\Delta y)}{\Delta x} \Delta x + \frac{z(x, y+\Delta y) - z(x, y)}{\Delta y} \Delta y$$

Taking the limits as $\Delta x \rightarrow 0$ and $\Delta y \rightarrow 0$ and using the definitions of partial derivatives, we obtain

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

is the fundamental relation for the **total differential** of a dependent variable in terms of its partial derivatives with respect to the independent variables. This relation can easily be extended to include more independent variables.

Consider air at 300 K and 0.86 m^3/kg. The state of air changes to 302 K and 0.87 m^3/kg as a result of some disturbance. Estimate the change in the pressure of air.

The temperature and specific volume of air changes slightly during a process. The resulting change in pressure is to be determined.

The changes in T and v , respectively, can be expressed as

$$dT \cong \Delta T = (302 - 300) = 2\text{K}$$

$$dv \cong \Delta v = (0.87 - 0.86) = 0.01\ \text{m}^3/\text{kg}$$

An ideal gas obeys the relation $pv = RT$. Solving for p yields

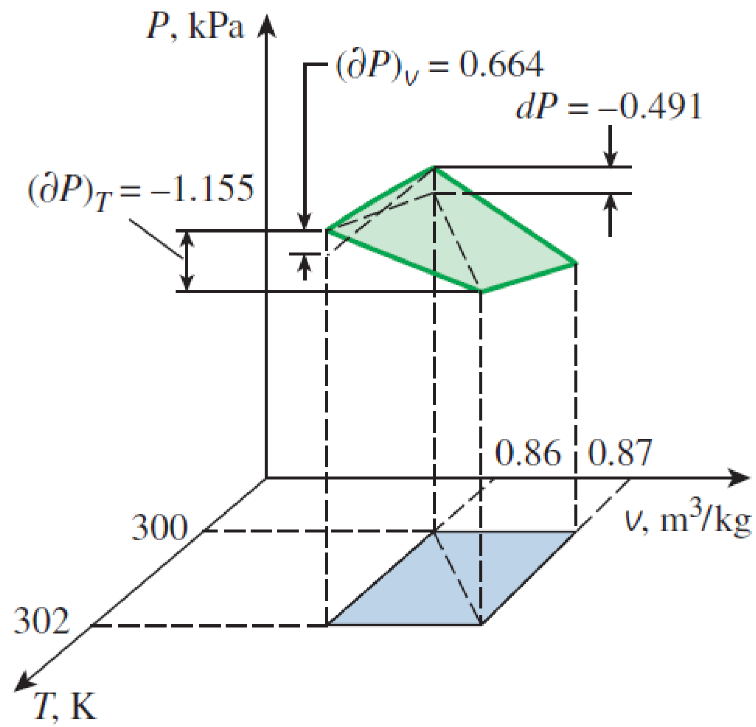
$$p = \frac{RT}{v}$$ As R is a constant $p = p(T, v)$

$$dp = \left(\frac{\partial p}{\partial T}\right)_v dT + \left(\frac{\partial p}{\partial v}\right)_T dv$$

$$dp = \frac{R}{v} dT - \frac{RT}{v^2} dv$$

$$= 0.287 \left[\frac{2}{0.865} - \frac{301 \times 0.01}{0.865^2}\right]$$

$$= -0.491$$



$$\left(\frac{\partial p}{\partial T}\right)_v dT = 0.664\ \text{kPa} \qquad \left(\frac{\partial p}{\partial v}\right)_T dv = -1.155\ \text{kPa}$$

Mathematical Theorems

There are two simple theorems in partial differential calculus that are used very often in this chapter. The proofs are as follows. Suppose there exists a functional relationship between three coordinates. x, y, and z; thus

$$f(x,y,z) = 0 \ldots..(1)$$

Then x can be imagined as a function of y and z and

$$dx = \left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial x}{\partial z}\right)_y dz \ldots (2).$$

Also y can be imagined as a function of x and z and

$$dy = \left(\frac{\partial y}{\partial x}\right)_z dx + \left(\frac{\partial y}{\partial z}\right)_x dz \ldots.(3)$$

Substituting 3 in 2

$$dx = \left(\frac{\partial x}{\partial y}\right)_z \left[\left(\frac{\partial y}{\partial x}\right)_z dx + \left(\frac{\partial y}{\partial z}\right)_x dz\right] + \left(\frac{\partial x}{\partial z}\right)_y dz \ldots (4).$$

$$dx = \left[\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial x}\right)_z\right] dx + \left[\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x + \left(\frac{\partial x}{\partial z}\right)_y\right] dz \ldots \ldots \ldots \ldots \ldots \ldots \ldots (5)$$

Of the three coordinates , only two are independent. Choosing x and z are the independent coordinates, the above equation must be true for all set of values of dx and dz. Thus if dz=0

$dx \neq 0$ Which results $\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial x}\right)_z = 1$

Reciprocity Relation $\left(\frac{\partial x}{\partial y}\right)_z = \frac{1}{\left(\frac{\partial y}{\partial x}\right)_z}$

$dx = 0$ and $dz \neq 0$ it follows that $\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x + \left(\frac{\partial x}{\partial z}\right)_y = 0$

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x = -\left(\frac{\partial x}{\partial z}\right)_y$$

Applying the reciprocity relation

Cyclic Relation $\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$

Function: $z + 2xy - 3y^2z = 0$

1) $z = \frac{2xy}{3y^2-1} \rightarrow \left(\frac{\partial z}{\partial x}\right)_y = \frac{2y}{3y^2-1}$

2) $x = \frac{3y^2z - z}{2y} \rightarrow \left(\frac{\partial x}{\partial z}\right)_y = \frac{3y^2-1}{2y}$

Thus, $\left(\frac{\partial z}{\partial x}\right)_y = \frac{1}{\left(\frac{\partial x}{\partial z}\right)_y}$

In case of a PVT system the 2nd theorem yields

$$\left(\frac{\partial p}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial T}{\partial p}\right)_V = -1$$

$$\left(\frac{\partial p}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial p}{\partial T}\right)_V$$

The volume expansion coefficient is defined as: $\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$

The isothermal compressibility is defined as: $\kappa = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$

Therefore $\left(\frac{\partial p}{\partial T}\right)_V = \frac{\beta}{\kappa}$ $\quad dp = \frac{\beta}{\kappa}dT - \frac{1}{\kappa V}dv$ At constant volume: $dp = \frac{\beta}{\kappa}dT$

For small change in temperature $p_f - p_i = \frac{\beta}{\kappa}(T_f - T_i)$

Internal Energy Changes

We choose the internal energy to be a function of T and v

$$u = u(T, v)$$

$$du = \left(\frac{\partial u}{\partial T}\right)_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv$$

Using the definition of c_v $\quad c_v = \left(\frac{\partial u}{\partial T}\right)_v \qquad du = c_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv$

For constant density (constant specific volume) substance (loosely incompressible) $dv = 0$ or for substances with specific internal energy independent of volume.

$$du = c_v dT$$

$$u_2 - u_1 = \int_{T_1}^{T_2} c_v dT$$

Enthalpy Changes

The general relation for dh is determined in exactly the same manner. This time we choose the enthalpy to be a function of T and P, that is, h=f (T, p) and take its total differential,

$$h = h(T, p)$$

$$dh = \left(\frac{\partial h}{\partial T}\right)_p dT + \left(\frac{\partial h}{\partial p}\right)_T dp \dots (a)$$

Using the definition of c_p, we have

$$dh = c_p\, dT + \left(\frac{\partial h}{\partial p}\right)_T dp \dots (b)$$

If the process is isobaric or if a substance has a specific enthalpy independent of pressure equation b reduces to

$$dh = c_p dT$$

$$h_2 - h_1 = \int_{T_1}^{T_2} c_p dT$$

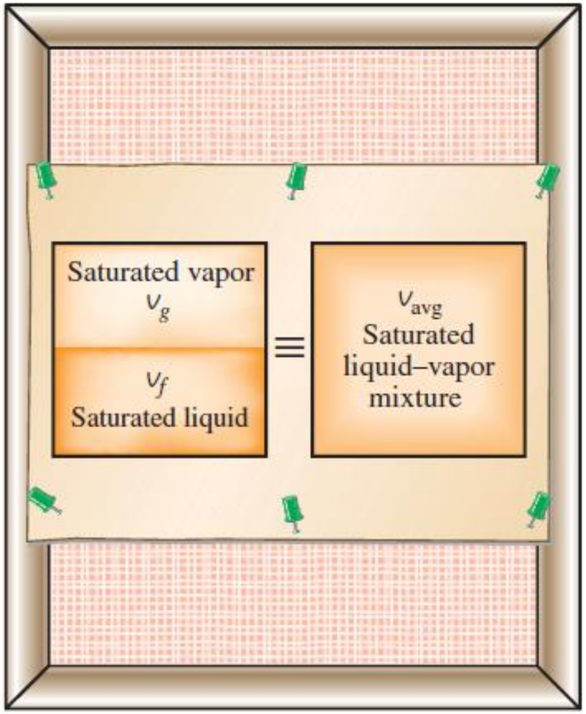
If the c_p is constant for averaged over the temperature range T_1 to T_2 then

$$h_2 - h_1 = c_p(T_2 - T_1)$$

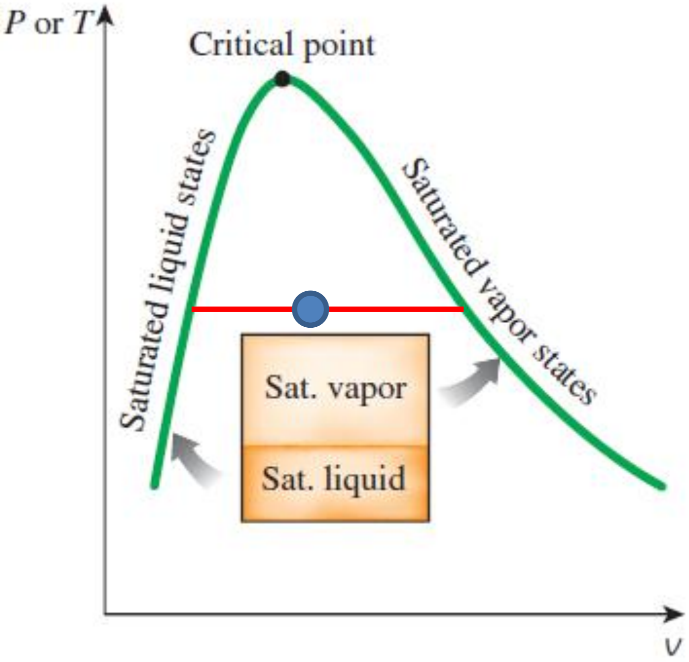
Evaluation of thermodynamic properties

In general thermodynamic properties can be evaluated from:

1. Thermodynamic equations of state (EOS).
2. Thermodynamic tables.
3. Thermodynamic charts.
4. Direct experimental results, and
5. The formulae of statistical thermodynamics



Properties of the Liquid-Vapor mixture



$$v = (1-x)v_f + xv_g = v_f + xv_{fg}$$
$$u = (1-x)u_f + xu_g = u_f + xu_{fg}$$
$$h = (1-x)h_f + xh_g = h_f + xh_{fg}$$
$$s = (1-x)s_f + xs_g = s_f + xs_{fg}$$
$$\frac{1}{\rho} = (1-x)\frac{1}{\rho_f} + x\frac{1}{\rho_g}$$

x is the Dryness fraction or Quality

Thermodynamic tables

Properties given in the problem statement	Choose	Look up in the appropriate table	Then you have a		
			Compressed liquid if	Mixture of liquid and vapour if	Superheated vapour if
p_{given} & T_{given}	p_{given}	T_{sat} at p_{given}	$T_{given} < T_{sat}$	$T_{given} = T_{sat}$	$T_{given} > T_{sat}$
p_{given} & T_{given}	T_{given}	p_{sat} at T_{given}	$p_{given} > p_{sat}$	$p_{given} = p_{sat}$	$p_{given} < p_{sat}$
p_{given} & v_{given}	p_{given}	v_f & v_g	$v_{given} < v_f$	$v_f < v_{given} < v_g$	$v_{given} > v_g$
p_{given} & u_{given}	p_{given}	u_f & u_g	$u_{given} < u_f$	$u_f < u_{given} < u_g$	$u_{given} > u_g$
p_{given} & h_{given}	p_{given}	h_f & h_g	$h_{given} < h_f$	$h_f < h_{given} < h_g$	$h_{given} > h_g$
T_{given} & v_{given}	T_{given}	v_f & v_g	$v_{given} < v_f$	$v_f < v_{given} < v_g$	$v_{given} > v_g$
T_{given} & u_{given}	T_{given}	u_f & u_g	$u_{given} < u_f$	$u_f < u_{given} < u_g$	$u_{given} > u_g$
T_{given} & h_{given}	T_{given}	h_f & h_g	$h_{given} < h_f$	$h_f < h_{given} < h_g$	$h_{given} > h_g$

Interpolation

If we wish to calculate v at a given pressure p, which does not appear explicitly in the table , we look in the table at p_1, which is below p, and p_2 that is just above it and find the corresponding values of v_1 and v_2. The value of v interpolated from

$$v = v_1 + \frac{p - p_1}{p_2 - p_1}(v_2 - v_1) = \frac{p_2 - p}{p_2 - p_1}v_1 + \frac{p - p_1}{p_2 - p_1}v_2$$

$$v = a_1 v_1 + a_2 v_2 = a_1 v_1 + (1 - a_1)v_2$$

If the specific v required as a function of known pressure p, and temperature T, whose values do not appear explicitly in the table double interpolation is used.

$$v = a_1 b_1 v_{11} + a_1 b_2 v_{12} + a_2 b_1 v_{21} + a_2 b_2 v_{22}$$

$$a_1 = \frac{p_2 - p}{p_2 - p_1} \qquad b_1 = \frac{T_2 - T}{T_2 - T_1}$$

$$a_2 = \frac{p - p_1}{p_2 - p_1} \qquad b_2 = \frac{T - T_1}{T_2 - T_1}$$

	p_1	p_2
T_1	v_{11}	v_{12}
T_2	v_{21}	v_{22}

Saturated Liquid water and Steam

T	P	v_f	v_g	v_{fg}	u_f	u_g	u_{fg}	pv_f	pv_g	$u_f + pv_f$	$u_g + pv_g$
ºC	(kPa)	m^3/kg	m^3/kg	m^3/kg	kJ/kg	kJ/kAg	kJ/kg	kJ/kg	kJ/kg	kJ/kg	kJ/kg
0.01	0.6113	0.001000	206.131	206.131	0	2375.33	2375.33	0.000611	126.0079	0.000611	2501.338
5	0.8721	0.001000	147.117	147.118	20.97	2382.24	2361.27	0.000872	128.3007	20.97087	2510.541
100	101.3	0.001044	1.67185	1.67290	418.19	2506.50	2087.58	0.105757	169.3584	418.2958	2675.858
190	1254.4	0.001141	0.15654	0.15539	806.17	2590.01	1783.84	1.43127	196.3638	807.6013	2786.374
250	3973	0.001251	0.05013	0.04887	1080.37	2602.37	1522.00	4.970223	199.1665	1085.34	2801.536
300	8581	0.001404	0.02167	0.02027	1331.97	2562.96	1230.99	12.04772	185.9503	1344.018	2748.91
374.1	22089	0.003155	0.003155	0	2029.58	2029.58	0	69.6908	69.6908	2099.271	2099.271

T	T	P	h_f	h_g	h_{fg}	s_f	s_g	s_{fg}	$g_f = h_f - Ts_f$	$g_g = h_g - Ts_g$
ºC	K	(kPa)	kJ/kg	kJ/kg	kJ/kg	kJ/kg-K	kJ/kg-K	kJ/kg-K	kJ/kg	kJ/kg
0.01	273.16	0.6113	0.00	2501.35	2501.35	0	9.1562	9.1562	0	0.242408
5	278.15	0.8721	20.98	2510.54	2489.57	0.0761	9.0257	8.9496	-0.18721	0.041545
100	373.15	101.3	419.02	2676.05	2257.03	1.3068	7.3548	6.0480	-68.6124	-68.3936
190	463.15	1254.4	807.61	2786.37	1978.76	2.2358	6.5078	4.2720	-227.901	-227.718
250	523.15	3973	1085.34	2801.52	1716.18	2.7927	6.0729	3.2802	-375.661	-375.518
300	573.15	8581	1344.01	2748.94	1404.93	3.2533	5.7044	2.4511	-520.619	-520.537
374.1	647.25	22089	2099.26	2099.26	0	4.4297	4.4297	0	-767.863	-767.863

TABLE B.1.4

Compressed Liquid Water

Temp. (°C)	v (m^3/kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg-K)	v (m^3/kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg-K)
	500 kPa (151.86°C)				2000 kPa (212.42°C)			
Sat.	0.001093	639.66	640.21	1.8606	0.001177	906.42	908.77	2.4473
0.01	0.000999	0.01	0.51	0.0000	0.000999	0.03	2.03	0.0001
20	0.001002	83.91	84.41	0.2965	0.001001	83.82	85.82	.2962
40	0.001008	167.47	167.98	0.5722	0.001007	167.29	169.30	.5716
60	0.001017	251.00	251.51	0.8308	0.001016	250.73	252.77	.8300
80	0.001029	334.73	335.24	1.0749	0.001028	334.38	336.44	1.0739
100	0.001043	418.80	419.32	1.3065	0.001043	418.36	420.45	1.3053
120	0.001060	503.37	503.90	1.5273	0.001059	502.84	504.96	1.5259
140	0.001080	588.66	589.20	1.7389	0.001079	588.02	590.18	1.7373
160	—	—	—	—	0.001101	674.14	676.34	1.9410
180	—	—	—	—	0.001127	761.46	763.71	2.1382
200	—	—	—	—	0.001156	850.30	852.61	2.3301

Saturated Table

T	P	v_f	u_f	h_f	s_f
°C	(kPa)	m^3/kg	kJ/kg	kJ/kg	kJ/kg-K
20	2.339	0.001002	83.94	83.94	0.2966
40	7.384	0.001008	167.53	167.54	0.5724
60	19.941	0.001017	251.09	251.11	0.8311
80	47.39	0.001029	334.84	334.88	1.0752
100	101.3	0.001044	418.91	419.02	1.3068
120	198.5	0.001060	503.48	503.69	1.5275
140	361.3	0.001080	588.72	589.11	1.7390

Compressed Liquid Table

	30000 kPa				50000 kPa			
0	0.000986	0.25	29.82	0.0001	0.000977	0.20	49.03	−0.0014
20	0.000989	82.16	111.82	0.2898	0.000980	80.98	130.00	0.2847
40	0.000995	164.01	193.87	0.5606	0.000987	161.84	211.20	0.5526
60	0.001004	246.03	276.16	0.8153	0.000996	242.96	292.77	0.8051
80	0.001016	328.28	358.75	1.0561	0.001007	324.32	374.68	1.0439
100	0.001029	410.76	441.63	1.2844	0.001020	405.86	456.87	1.2703
120	0.001044	493.58	524.91	1.5017	0.001035	487.63	539.37	1.4857
140	0.001062	576.86	608.73	1.7097	0.001052	569.76	622.33	1.6915
160	0.001082	660.81	693.27	1.9095	0.001070	652.39	705.91	1.8890
180	0.001105	745.57	778.71	2.1024	0.001091	735.68	790.24	2.0793
200	0.001130	831.34	865.24	2.2892	0.001115	819.73	875.46	2.2634
220	0.001159	918.32	953.09	2.4710	0.001141	904.67	961.71	2.4419
240	0.001192	1006.84	1042.60	2.6489	0.001170	990.69	1049.20	2.6158
260	0.001230	1097.38	1134.29	2.8242	0.001203	1078.06	1138.23	2.7860
280	0.001275	1190.69	1228.96	2.9985	0.001242	1167.19	1229.26	2.9536
300	0.001330	1287.89	1327.80	3.1740	0.001286	1258.66	1322.95	3.1200
320	0.001400	1390.64	1432.63	3.3538	0.001339	1353.23	1420.17	3.2867
340	0.001492	1501.71	1546.47	3.5425	0.001403	1451.91	1522.07	3.4556
360	0.001627	1626.57	1675.36	3.7492	0.001484	1555.97	1630.16	3.6290
380	0.001869	1781.35	1837.43	4.0010	0.001588	1667.13	1746.54	3.8100

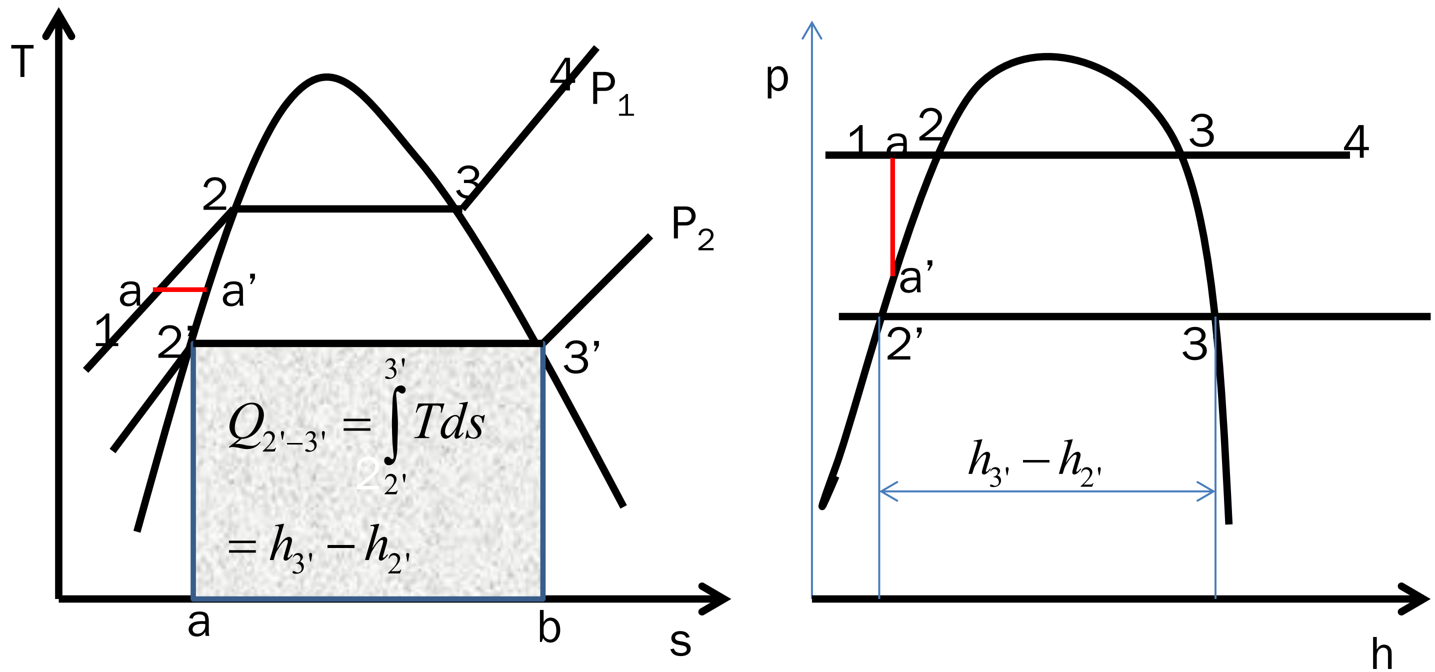
Saturated Table

T	P	v_f	u_f	h_f	s_f
°C	(kPa)	m³/kg	kJ/kg	kJ/kg	kJ/kg-K
20	2.339	0.001002	83.94	83.94	0.2966
40	7.384	0.001008	167.53	167.54	0.5724
60	19.941	0.001017	251.09	251.11	0.8311
80	47.39	0.001029	334.84	334.88	1.0752
100	101.3	0.001044	418.91	419.02	1.3068
120	198.5	0.001060	503.48	503.69	1.5275
140	361.3	0.001080	588.72	589.11	1.7390

Superheated Vapor Water

Temp. (°C)	*v* (m³/kg)	*u* (kJ/kg)	*h* (kJ/kg)	*s* (kJ/kg-K)	*v* (m³/kg)	*u* (kJ/kg)	*h* (kJ/kg)	*s* (kJ/kg-K)
	$P = 10$ kPa (45.81°C)				$P = 50$ kPa (81.33°C)			
Sat.	14.67355	2437.89	2584.63	8.1501	3.24034	2483.85	2645.87	7.5939
50	14.86920	2443.87	2592.56	8.1749	—	—	—	—
100	17.19561	2515.50	2687.46	8.4479	3.41833	2511.61	2682.52	7.6947
150	19.51251	2587.86	2782.99	8.6881	3.88937	2585.61	2780.08	7.9400
200	21.82507	2661.27	2879.52	8.9037	4.35595	2659.85	2877.64	8.1579
250	24.13559	2735.95	2977.31	9.1002	4.82045	2734.97	2975.99	8.3555
300	26.44508	2812.06	3076.51	9.2812	5.28391	2811.33	3075.52	8.5372
400	31.06252	2968.89	3279.51	9.6076	6.20929	2968.43	3278.89	8.8641
500	35.67896	3132.26	3489.05	9.8977	7.13364	3131.94	3488.62	9.1545
600	40.29488	3302.45	3705.40	10.1608	8.05748	3302.22	3705.10	9.4177
700	44.91052	3479.63	3928.73	10.4028	8.98104	3479.45	3928.51	9.6599
800	49.52599	3663.84	4159.10	10.6281	9.90444	3663.70	4158.92	9.8852
900	54.14137	3855.03	4396.44	10.8395	10.82773	3854.91	4396.30	10.0967
1000	58.75669	4053.01	4640.58	11.0392	11.75097	4052.91	4640.46	10.2964
1100	63.37198	4257.47	4891.19	11.2287	12.67418	4257.37	4891.08	10.4858
1200	67.98724	4467.91	5147.78	11.4090	13.59737	4467.82	5147.69	10.6662
1300	72.60250	4683.68	5409.70	14.5810	14.52054	4683.58	5409.61	10.8382

T-s and p-h Diagrams for Liquid-Vapor regime



$u_a = u_{a'}$ $\quad$ $s_a = s_{a'}$ $\quad$ $h_a = h_{a'}$

In many cases the property data for the subcooled zone is not available, hence the property of the subcooled fluid can be assumed to be same as that of the saturated fluid at the same temperature, this approximation is based on the assumption that for all practical purposes the constant pressure lines are assumed to be coincident with saturated liquid line in the sub-cooled region.

To be more accurate (liquid is almost incompressible) $\quad h_a = h_{a'} + v(p_a - p_{a'})$

In the subcooled zone isotherms almost coincide with isentropes as $\quad s_a - s_{a'} = c\ln\frac{T_a}{T_{a'}} = 0 \Rightarrow s_a = s_{a'}$

Incompressible substance model

Since the specific internal energy of a substance modelled as incompressible depends only on the temperature. The specific heat cv is also a function of temperature alone.

$$c_v(T) = \frac{du}{dT}$$ Incompressible (constant density)

Enthalpy varies with both temperature and pressure.

$$h(T,p) = u(T) + pv$$ Incompressible (constant density)

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p = \frac{du}{dT} = c_v \qquad c_p = c_v = c$$

Change in specific internal energy and specific enthalpy between two states are given as:

$$u_2 - u_1 = \int_{T_1}^{T_2} c(T)dT$$

Incompressible (constant density)

$$h_2 - h_1 = u_2 - u_1 + v(p_2 - p_1) = \int_{T_1}^{T_2} c(T)dT + v(p_2 - p_1)$$

Incompressible (constant density)+ constant specific heat c

$$u_2 - u_1 = c(T_2 - T_1) \qquad h_2 - h_1 = c(T_2 - T_1) + v(p_2 - p_1)$$

Approximation for liquids using saturated liquid data

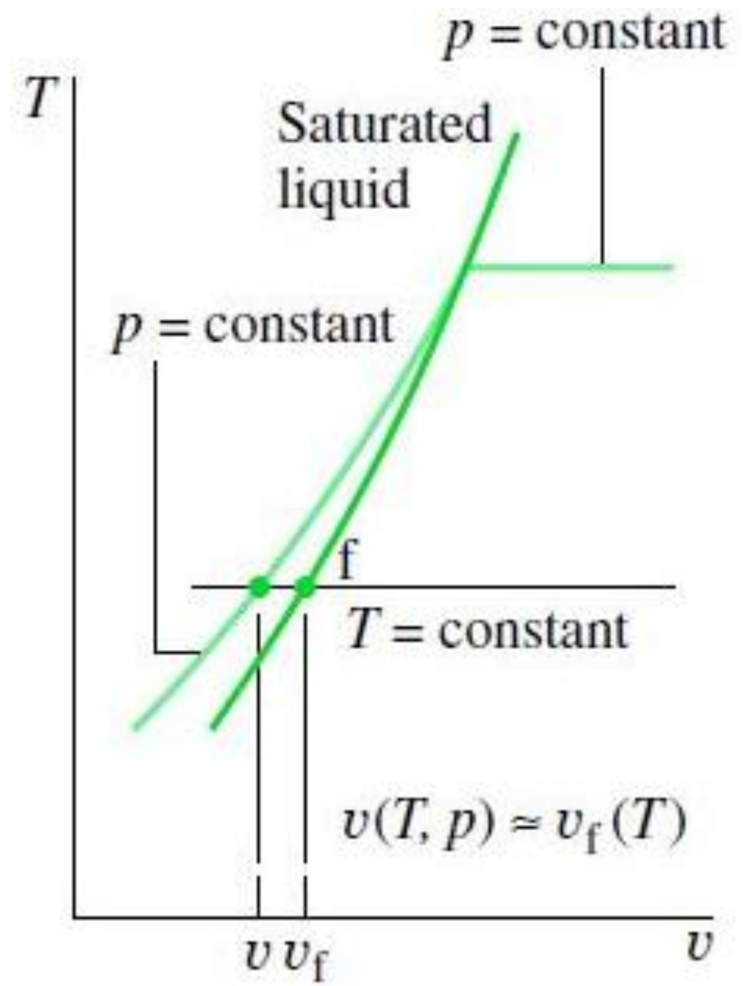
Because the values of v and u vary very little as pressure changes at fixed temperature, the following approximations are reasonable for many engineering calculation.

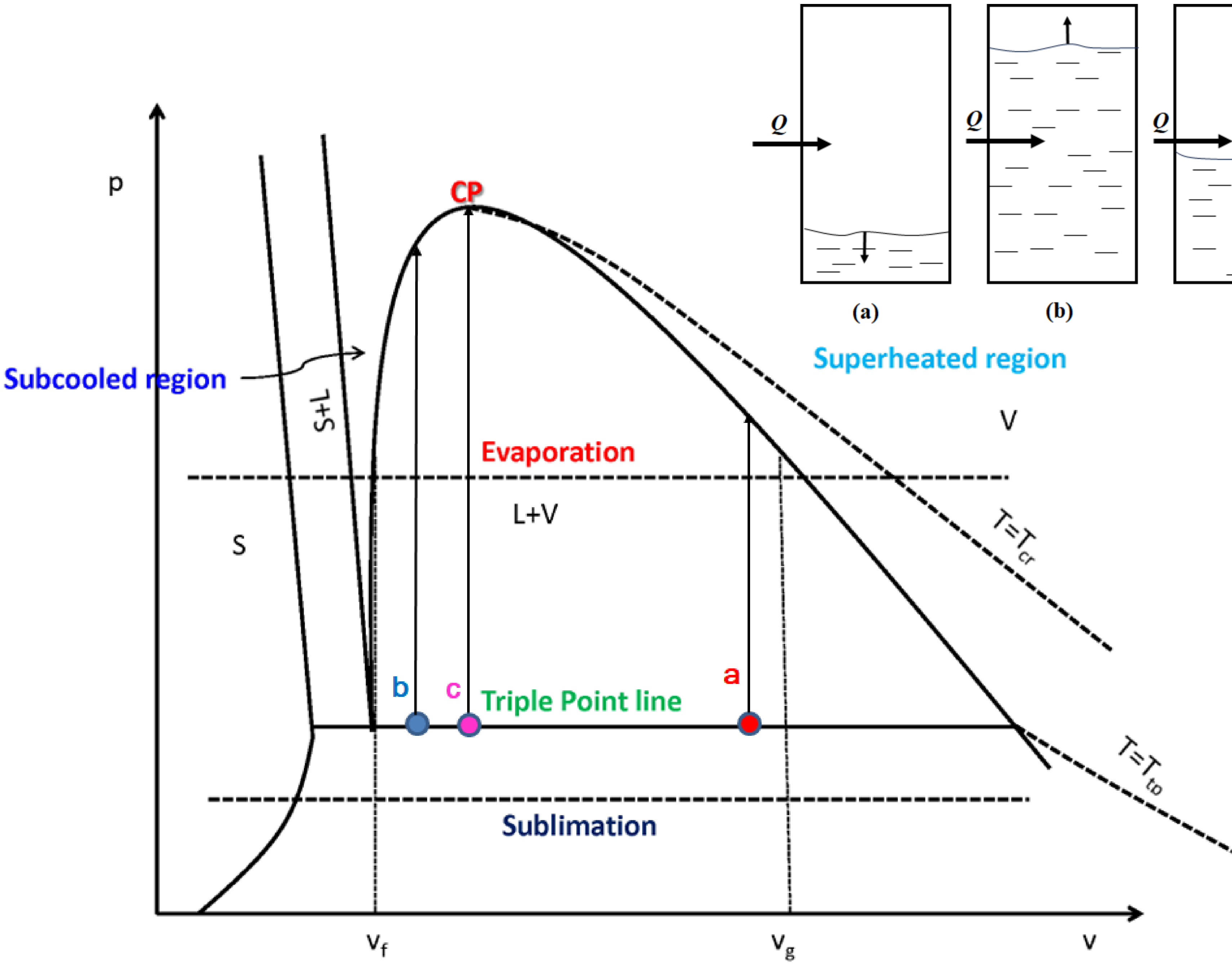
$$v(T,p) \approx v_f(T)$$
$$u(T,p) \approx u_f(T)$$

An approximate value of h :

$$h(T,p) \approx u_f(T) + pv_f(T) = u_f(T) + p_{sat}(T)v_f(T) - p_{sat}(T)v_f(T) + pv_f(T) = h_f(T) + v_f(T)(p - p_{sat}(T))$$

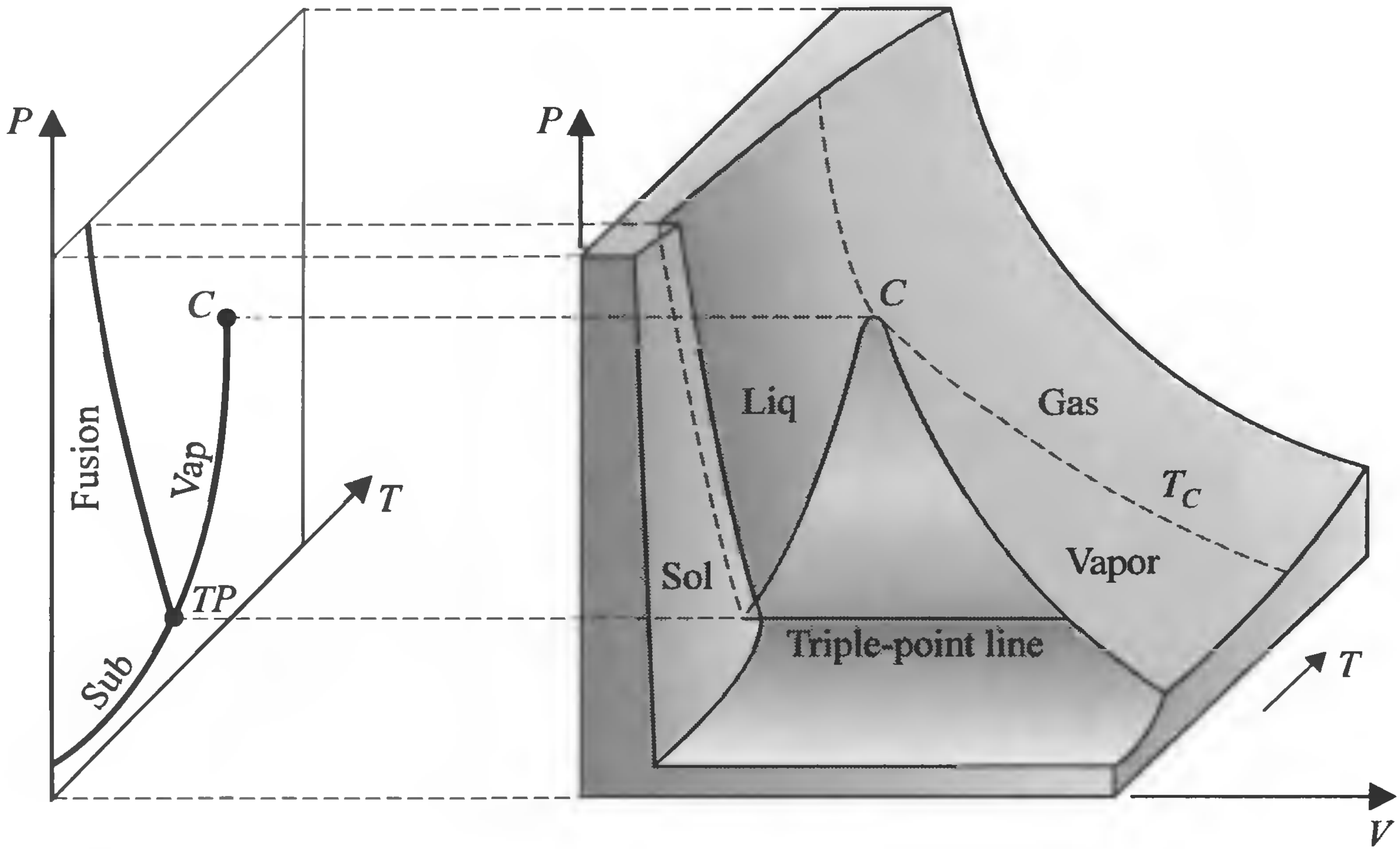
$$h(T,p) \approx h_f(T)$$





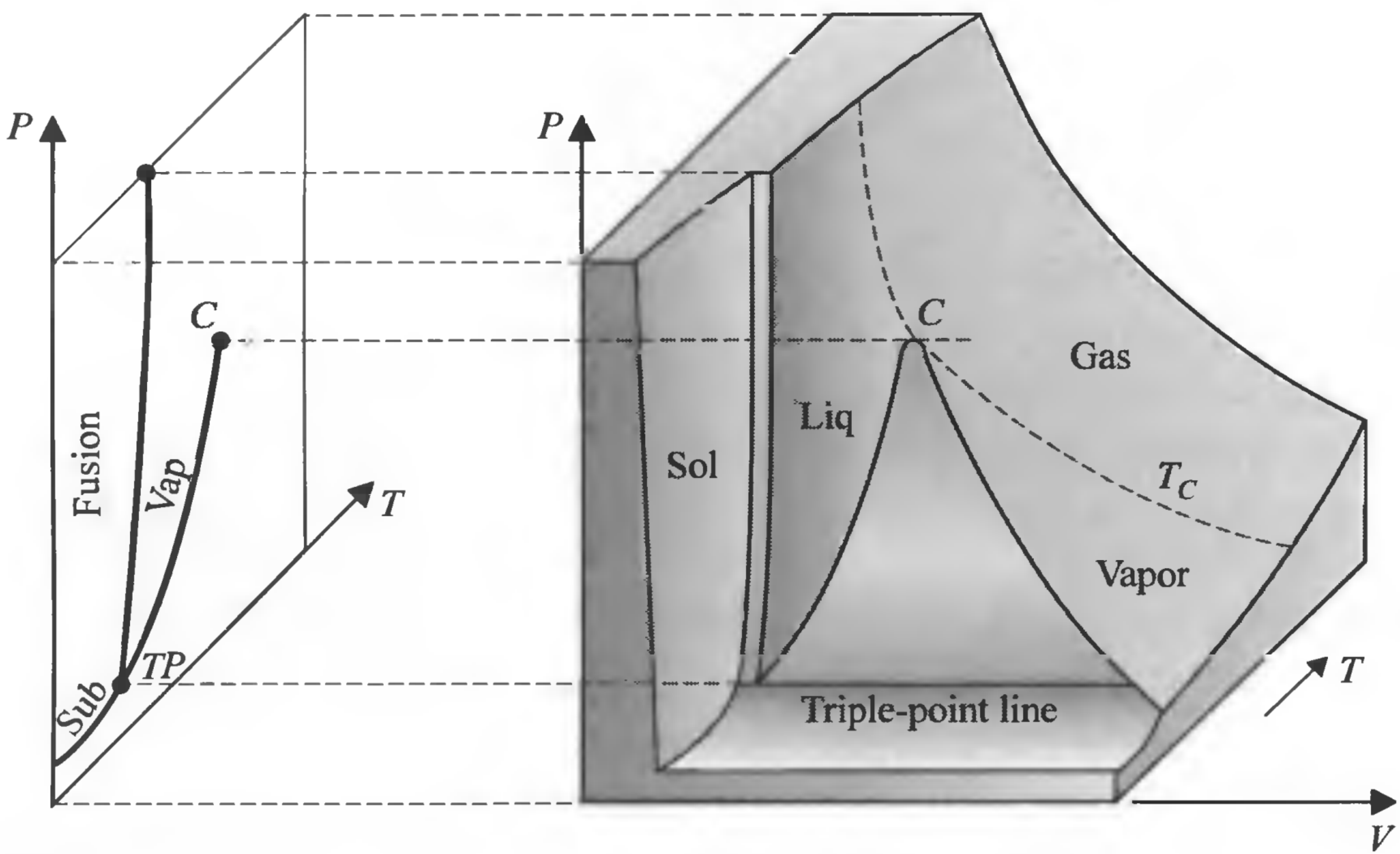
p
CP
Subcooled region
S+L
S
Evaporation
L+V
Superheated region
V
$T=T_{cr}$
b
c
Triple Point line
a
$T=T_{tp}$
Sublimation
v_f
v_g
v
Q
Q
Q
(a)
(b)
(c)

Thermodynamic Surfaces

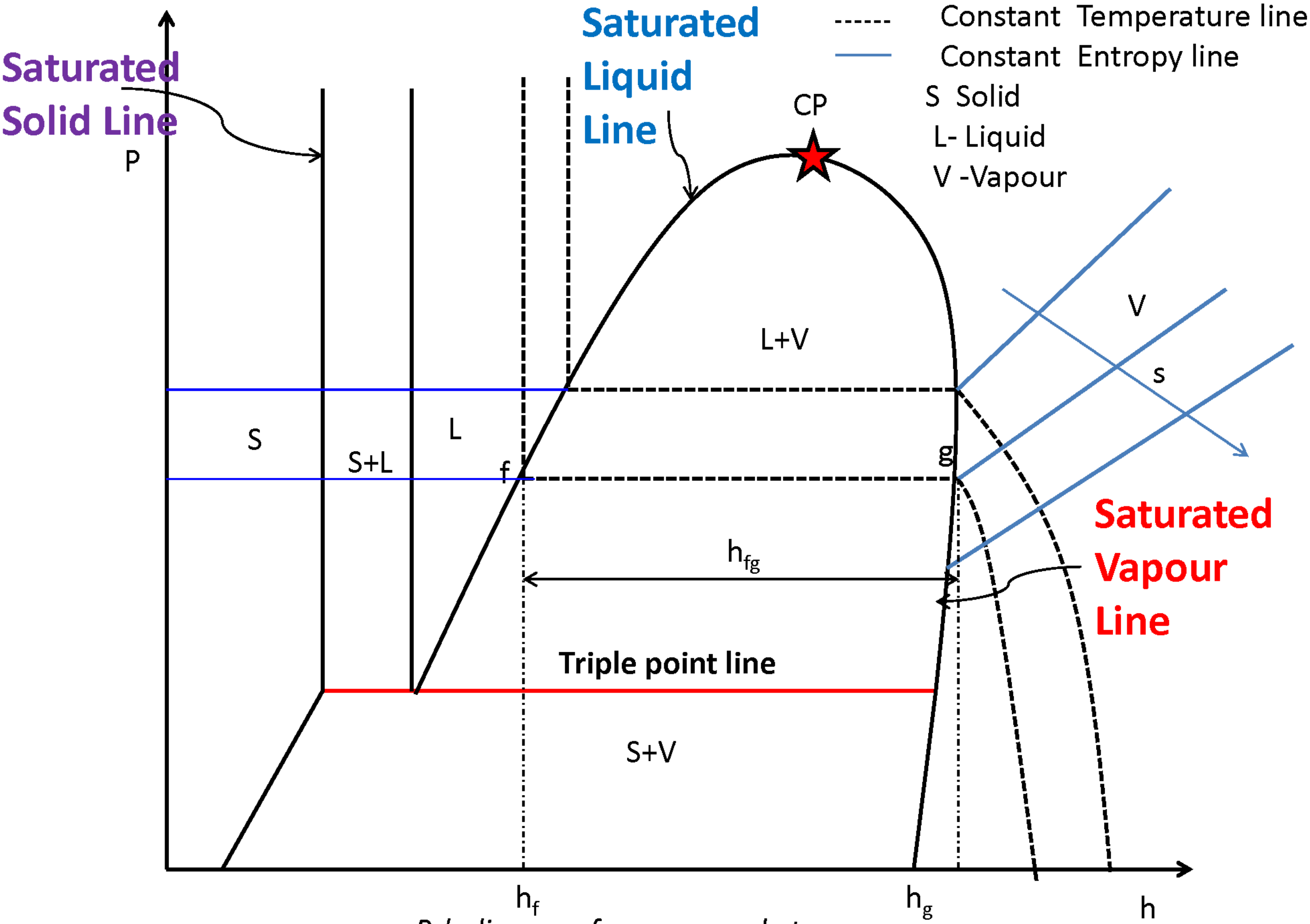


P-v-T surface of water which contracts while melting

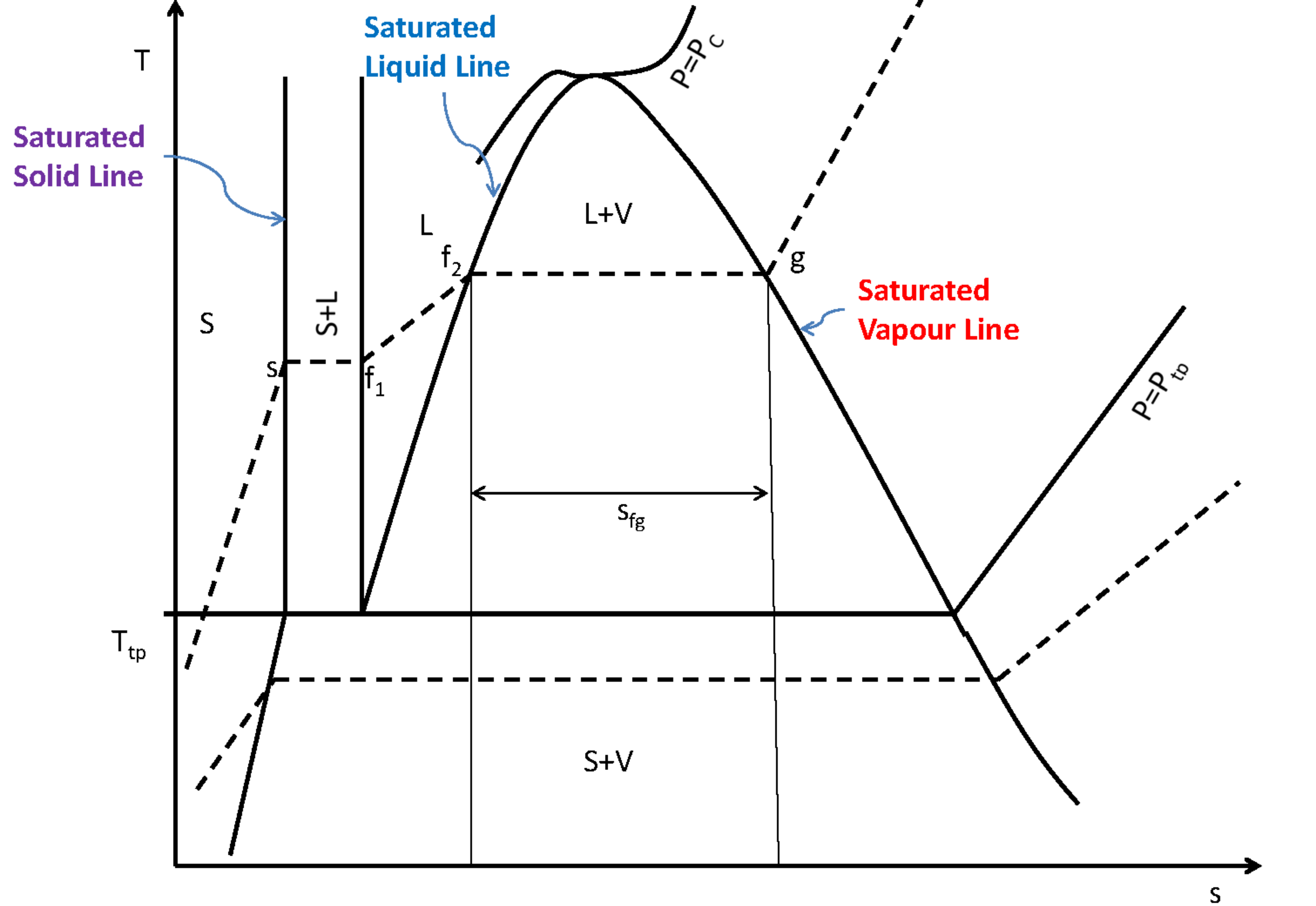
Thermodynamic Surfaces



P-v-T surface of CO_2 which expands while melting

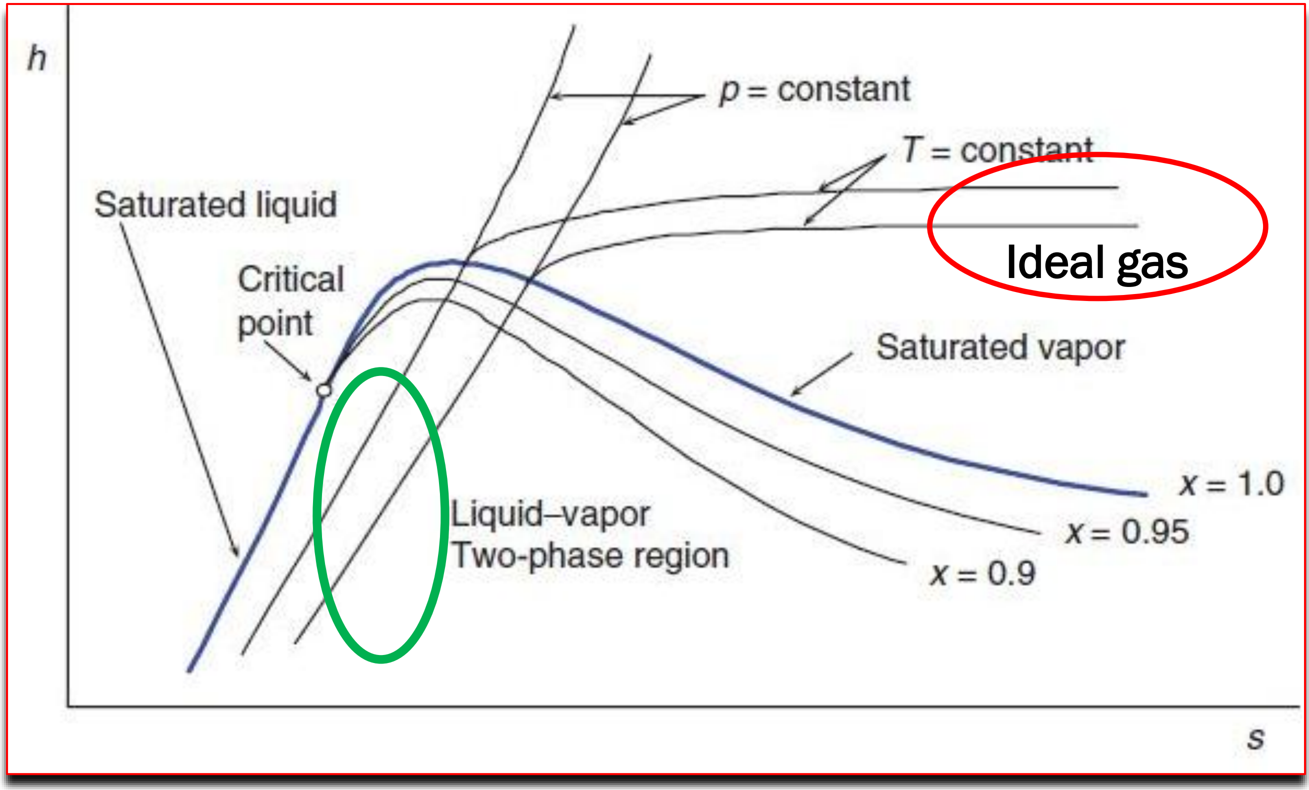


P-h diagram for a pure substance



T-s diagram for a pure substance

Mollier diagram.

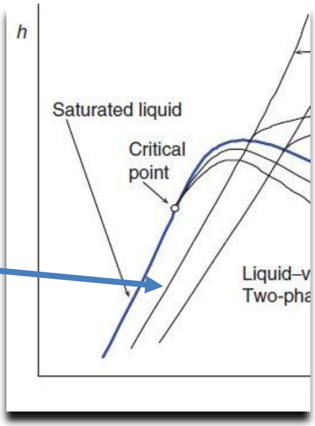


- The work of an adiabatic steady–state flow process is given by the vertical distance between the end points of the process, The enthalpy difference in a constant pressure process can be directly read from the h-s diagram which is the heat interaction associated with the process. The isentropic compression and expansion processes (ideal processes for compressor, pump and turbines) can be easily identified as the vertical lines.

- For an ideal gas, the $h-s$ diagram has the same shape as the $T-s$ diagram, because for an ideal gas the enthalpy is proportional to the temperature, $h = c_p T + \text{constant}$

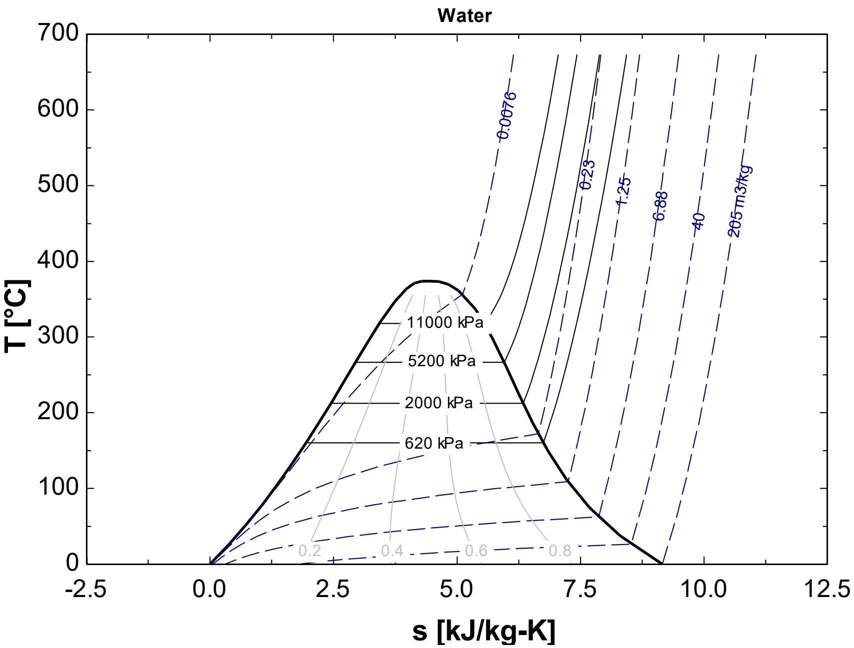
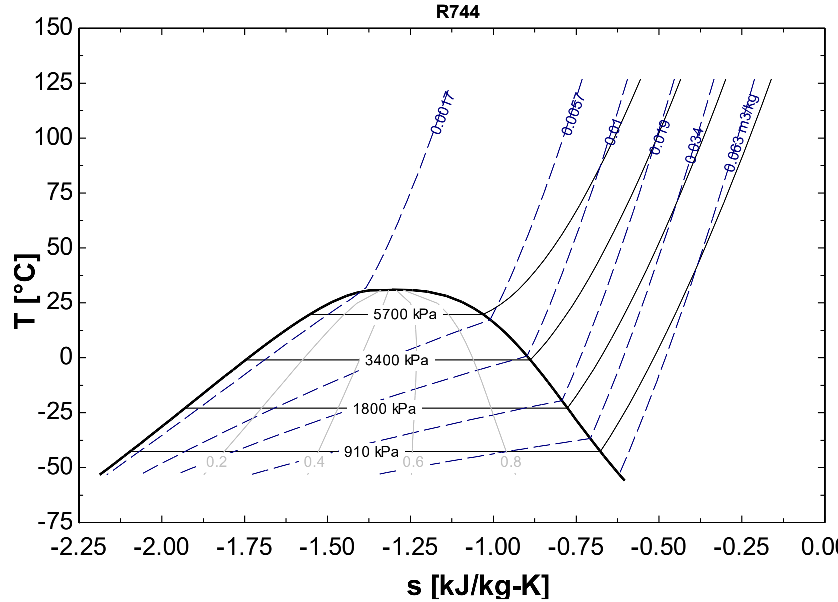
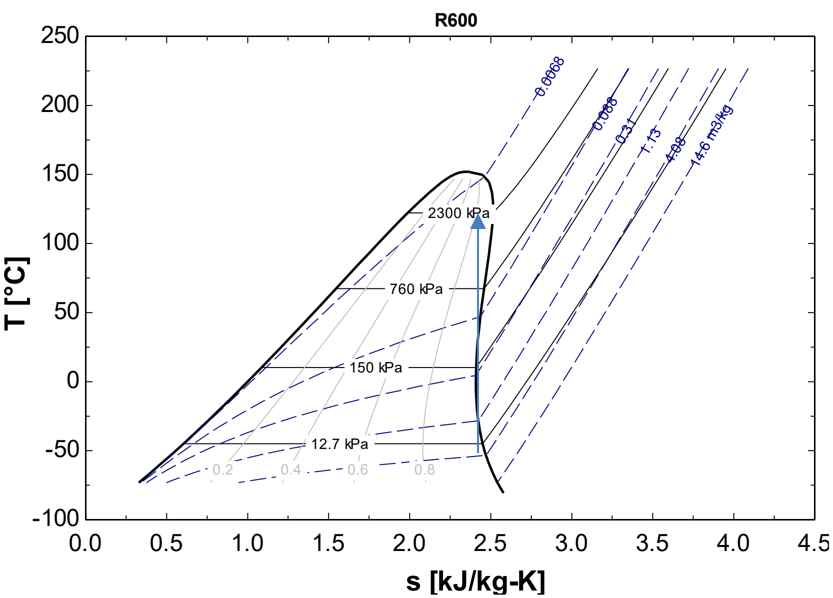
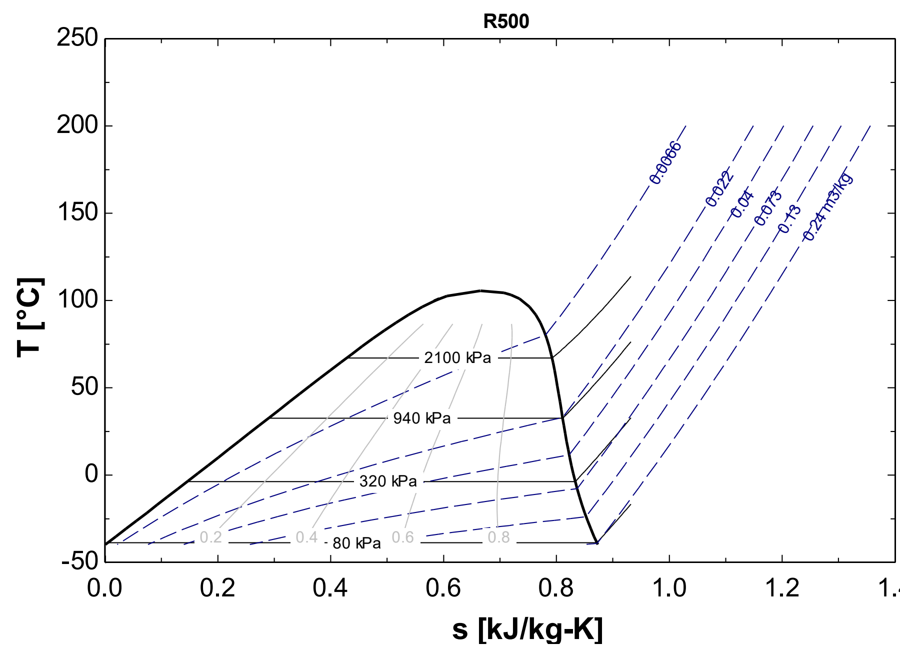
- Here also, the saturation curve is bell-shaped. However, it is skewed as compared with that of the $T-s$ diagram, and the critical point is not located at the highest point of the bell. At the critical point p and T are maximum but not the v hence h is not the maximum
- The constant-pressure lines in the two-phase region are also constant-temperature lines. These are straight lines and their slopes are given by their temperature, as $(\partial h/\partial s)p = T$. In the 2-phase region the pressure and temperature are dependent, hence constant pressure line coincide with constant temperature line. Thus the slope of the isobars remain constant in the two-phase region

Constant Pressure lines are straight lines within 2-phase region

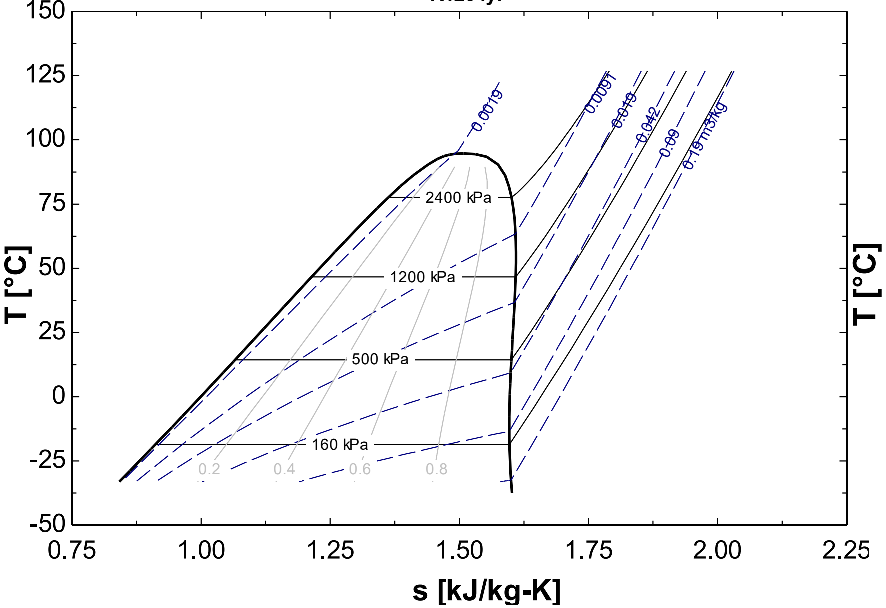


- In the superheated region, the constant-pressure lines approach exponential curves as the temperature increases, and the constant-temperature lines tend asymptotically to the horizontal. For ideal gas enthalpy is only a function of temperature.

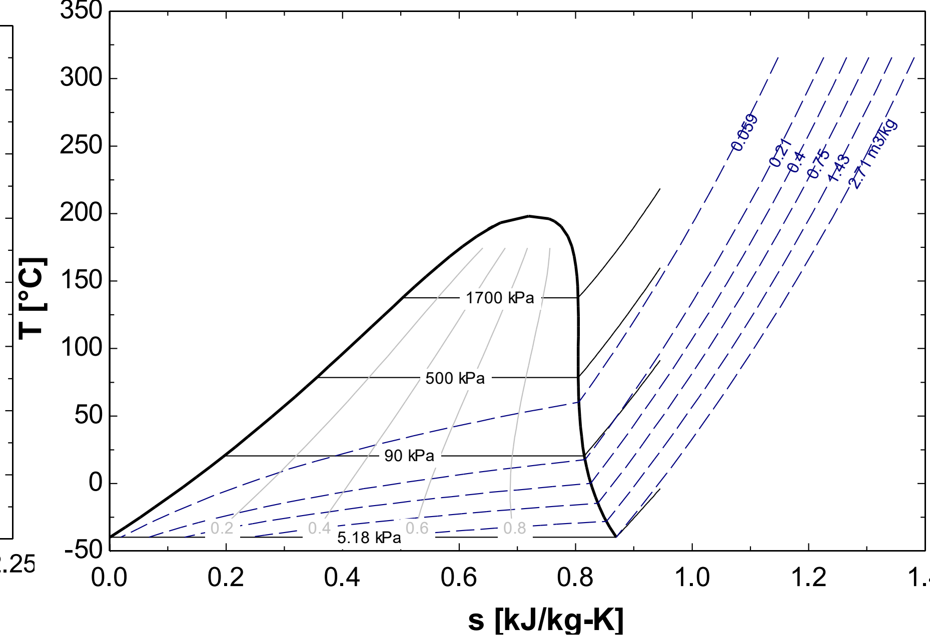
Note: $Tds = dh - vdp \Rightarrow \left(\frac{\partial h}{\partial s}\right)_p = T$


Water
T [°C]
s [kJ/kg-K]
700
600
500
400
300
200
100
0
-2.5
0.0
2.5
5.0
7.5
10.0
12.5
11000 kPa
5200 kPa
2000 kPa
620 kPa
0.0076
0.23
1.25
6.88
40
205 m3/kg
0.2
0.4
0.6
0.8

R744
T [°C]
s [kJ/kg-K]
150
125
100
75
50
25
0
-25
-50
-75
-2.25
-2.00
-1.75
-1.50
-1.25
-1.00
-0.75
-0.50
-0.25
0.00
5700 kPa
3400 kPa
1800 kPa
910 kPa
0.0017
0.0057
0.01
0.019
0.034
0.063 m3/kg
0.2
0.4
0.6
0.8

R600
T [°C]
s [kJ/kg-K]
250
200
150
100
50
0
-50
-100
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
2300 kPa
760 kPa
150 kPa
12.7 kPa
0.0068
0.088
0.31
1.13
4.08
14.6 m3/kg
0.2
0.4
0.6
0.8

R500
T [°C]
s [kJ/kg-K]
250
200
150
100
50
0
-50
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
2100 kPa
940 kPa
320 kPa
80 kPa
0.0066
0.022
0.04
0.073
0.13
0.24 m3/kg
0.2
0.4
0.6
0.8

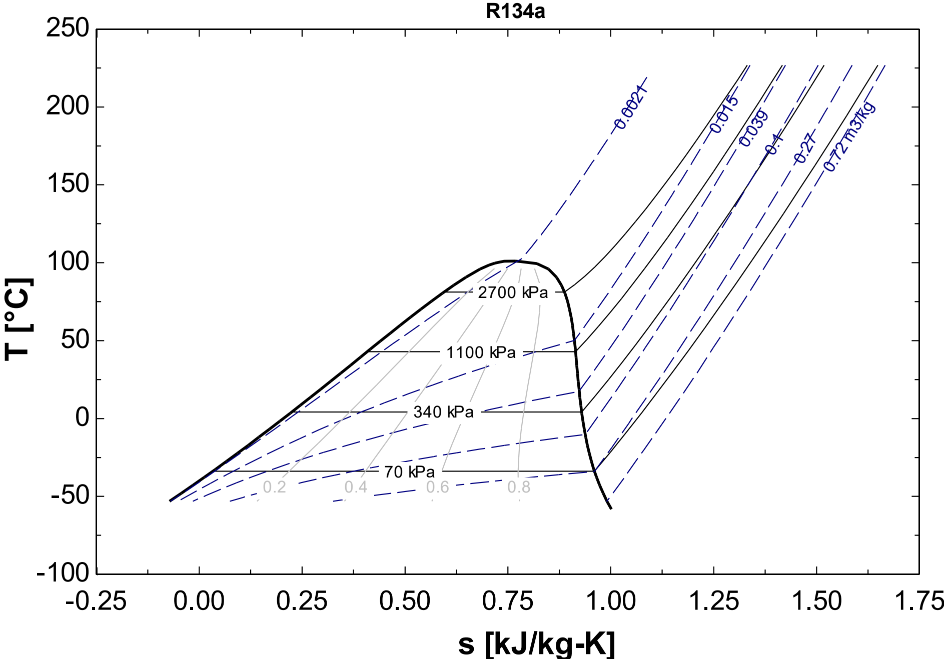
R1234yf



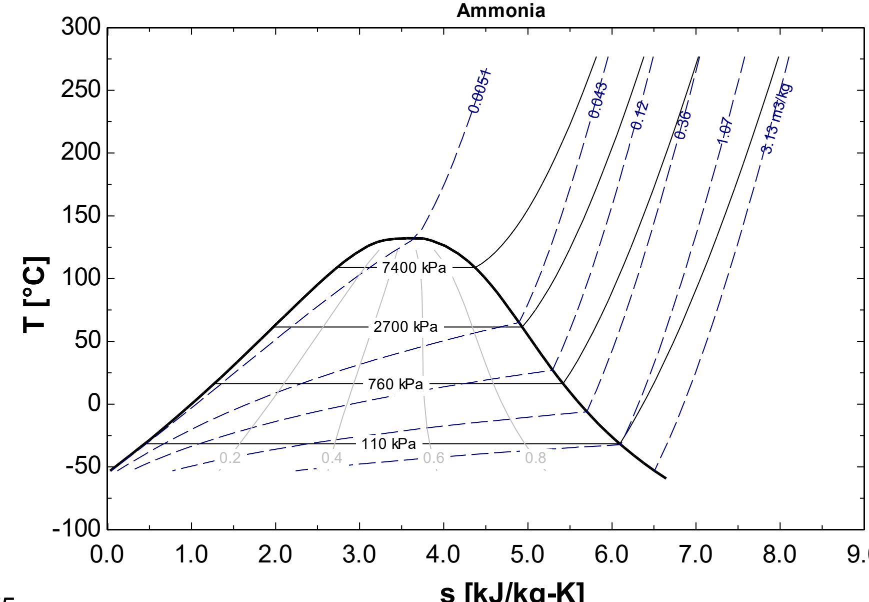
R11



R134a



Ammonia



Thermodynamic Equation of State

An equation of state relates the dependent properties with independent properties. For example,

$$f(P, v, T) = 0 \; (Mechanical\ Equation\ of\ State)$$

There are several EOSs, Some of them are very simple and others are very complicated (as many as 30 terms)

The simplest set of equation is that of ideal gas

Equation of State of Gases:

Thermodynamic equation of state: is a fundamental relation which express a relation between pressure, specific volume and temperature

Ideal/perfect gas: The inter molecular forces of attraction and repulsion between gas molecules are negligible and the actual volume of the molecules should be negligible compared to total volume for a perfect gas. Ideal gas obeys the equation of state given below at all temperature and pressure.

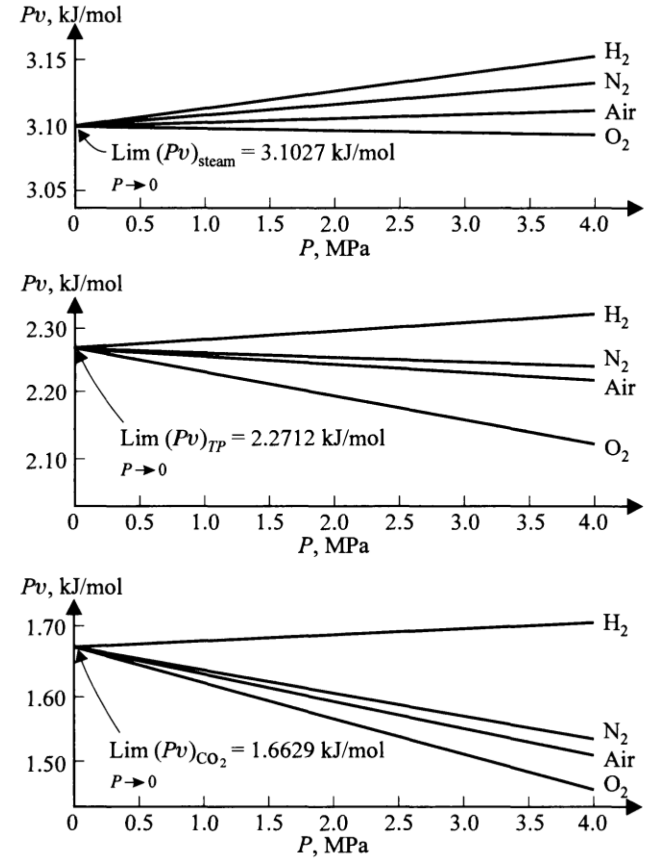
Ideal gas EOS

$$pV = mRT = m\frac{\bar{R}}{M}T = \frac{m}{M}\bar{R}T = n\bar{R}T \;;\; pv = RT;\, p = \rho RT;\, p\bar{v} = \bar{R}T$$

$$\left[\begin{array}{l} M - \dfrac{g}{gmole}\; or\; \dfrac{kg}{kmole})\; gmole = 6.023\times 10^{23};\, kmole = 6.023\times 10^{26} \\[2ex] \bar{R} - \dfrac{kJ}{kmole-K}\; or\; \dfrac{J}{gmole-K};\, R - \dfrac{kJ}{kg-K}\; or\; \dfrac{J}{g-K} \\[2ex] \bar{v} = \dfrac{V}{n} - molal\; specific\; volume\quad \dfrac{m^3}{kmole}\quad v = \dfrac{V}{m} = \dfrac{1}{\rho} - specific volume \end{array}\right]$$

Fundamental Property of Gases

The remarkable property of gases that makes them so valuable in thermometry is displayed in the figure. Where the product pv is plotted against p for four different gases all at the temperature of boiling water in the top graph, all at the triple point of water in the middle graph, and all at the temperature of solid CO_2 at the bottom graph. In each case it is seen that as pressure approaches zero, the product pv approaches same value for all gases at the same temperature.



$$\lim_{p\to 0}(pv) = A = \begin{Bmatrix} \text{Function of temperature only} \\ \text{independent of the gas} \end{Bmatrix}$$

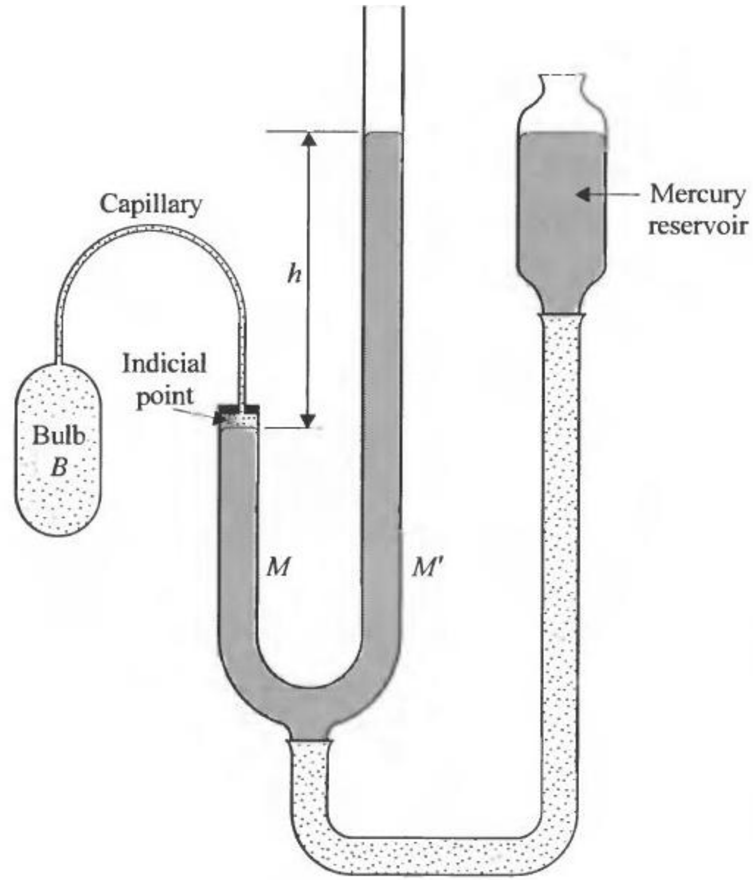
$$T = 273.16K \lim_{p_{tp}\to 0}\left(\frac{p}{p_{tp}}\right) \qquad (Const.V)$$

$$T = 273.16K lim\frac{pV/n}{p_{tp}V/n} = 273.16\frac{lim(pv)}{lim(p_{tp}v)} \qquad lim(pv) = \left[\frac{lim(p_{tp}v)}{273.16}\right]T = \bar{R}T$$

$$\bar{R} = \left[\frac{lim(p_{tp}v)}{273.16}\right] = \frac{2.2712\times 1000}{273.16} = 8.314\ kJ/kmol$$

Constant Volume Gas Thermometer

- The gas is contained in the glass bulb B, which communicates with mercury column M through a capillary.
- The volume of the gas is kept constant by adjusting the height of the mercury column M until the mercury level just touches the tip of the small pointer (indicial point)
- The mercury column M is adjusted by raising or lowering the reservoir.
- The pressure in the system equals atmospheric pressure plus the difference in height h between the two mercury columns M and M' and is measured twice : when the bulb is surrounded by the system whose temperature is to be measured , and then it is surrounded by water at triple point.



$$\theta(X) = aX \ (constant\ Y)$$

- As the coordinate X approaches zero, the temperature also approaches zero. In effect the linear function also defines an absolute temperature scale, such as Kelvin scale or the Rankine scale.

- It is well known that a gas is a best-behaved thermometric substance because of the fact that the ratio of the pressure p of a gas at any temperature to the pressure of the same gas at the triple point (p_{TP}) of water as both p and p_{TP} approaches zero, approaches a value independent of the nature of the gas . The limiting value of this ratio when multiplied by 273.16 K, was defined to be the ideal gas temperature T of the system at whose temperature the gas exert a pressure p. The reason for this regular behaviour may be found by investigating the way in which the product pv of a gas depends on p

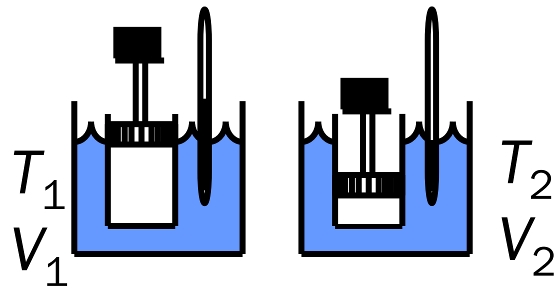
$$\boldsymbol{\theta(X) = aX\ (constant\ Y)}$$

$$\boldsymbol{\theta = 273.16K\ \frac{P}{P_{TP}}}$$

$$\boldsymbol{T = 273.16K} \lim_{P_{TP\to 0}} \boldsymbol{\frac{P}{P_{TP}}} \text{ (constant V)}$$

Important Gas Laws

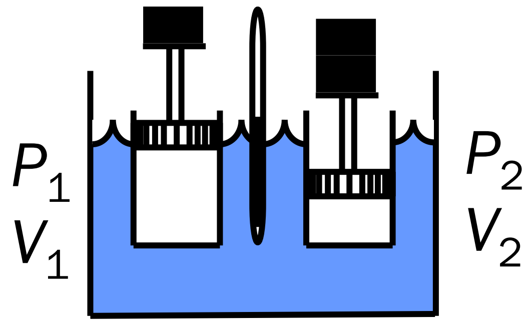
1. Charles' Law



P = const n = const

$$\frac{V_2}{V_1} = \frac{T_2}{T_1}$$

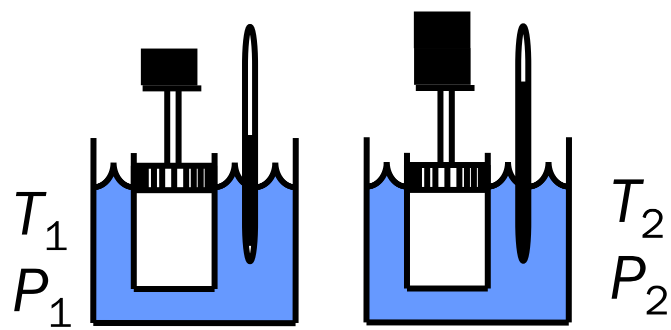
2. Boyle's Law



T = const n = const

$$\frac{P_2}{P_1} = \frac{V_1}{V_2}$$

3. Gay-Lussac's Law



V = const n = const

$$\frac{P_2}{P_1} = \frac{T_2}{T_1}$$

Avogadro's Law: Equal volume of ideal gases at the same pressure and temperature , contains the same number of molecules

$$\frac{V}{n} = k$$

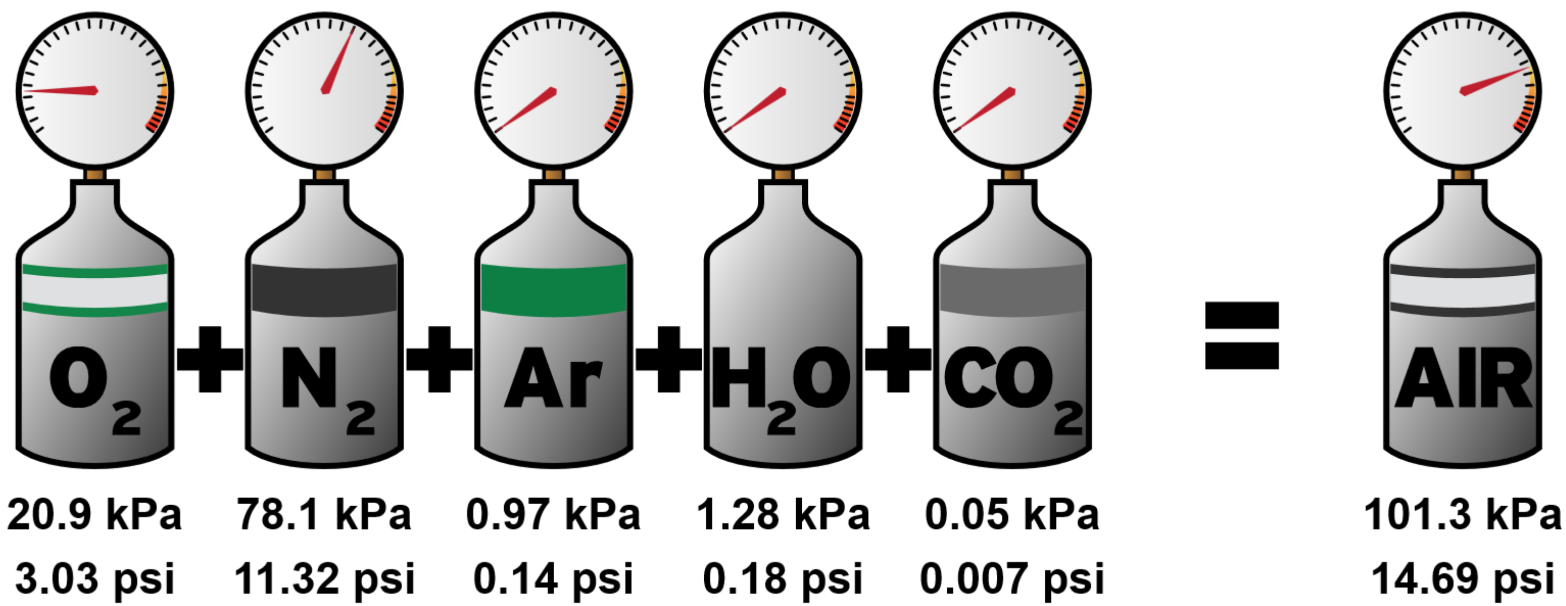
k is the molar volume

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

Under standard conditions for temperature and pressure, the value of T corresponds to 273.15 Kelvin and the value of P corresponds to 101.325 kilo Pascals. Therefore, the volume occupied by one mole (g mole) of a gas at STP is:

$$8.314 \times \frac{273.15}{101.325} = 22.4\ litres$$

Dalton's Law: The pressure of a mixture of ideal gases is equal to the sum of the partial pressure of each gas. Indirectly Dalton's law states that each constituent of an ideal gas mixture occupy the total volume of the container at the mixture temperature.



When does a real gas approache an ideal gas behaviour?

When the gas molecules are far apart from each other (large volume/ lower density)then the intermolecular force of attraction/repulsion is negligible, This situation happens when a large space is filled with very small number of gas molecules in other words under a low pressure condition **(compared to the critical pressure)**, Similarly if the gas **temperature is very high** (**compared to the critical temperature**) the molecules will be far apart, then actual volume of the molecules is negligible compared to the volume occupied. So from the above discussion it is concluded that a real gas approaches ideal gas behaviour as:

1. Molar mass decreases
2. Pressure decreases
3. Temperature increase

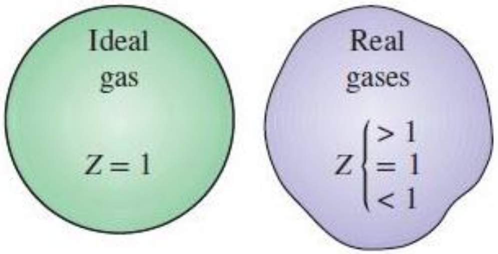
Inter Molecular Potential Energy may be Negligible

Note: In air-conditioning the water vapor in air may be considered as ideal gas as partial pressure of water vapor is lower than 10 kPa. Moist air can be treated as ideal gas till the pressure is 3 bar.

Compressibility Factor, Z
(A measure of deviation from ideal gas behaviour)

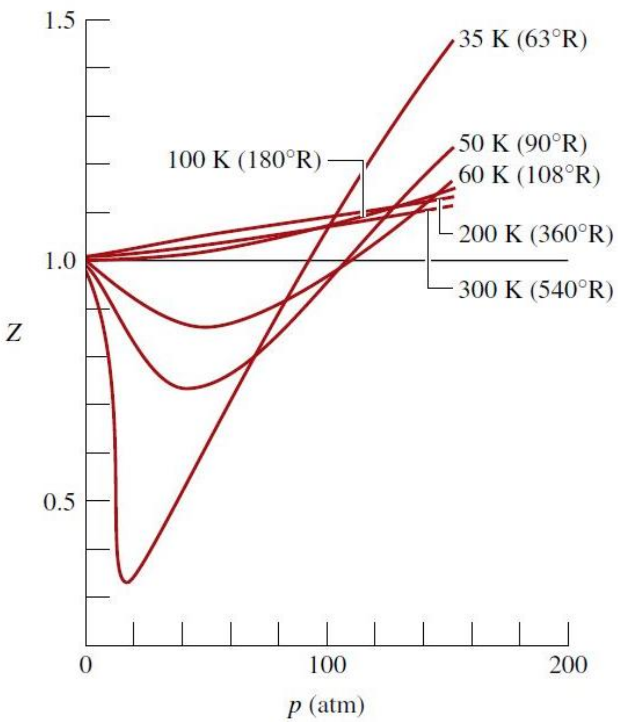
The dimensionless ratio $p\bar{v}/\bar{R}T$ or pv/RT is called the compressibility factor and is denoted by Z. That is,

$$Z = \frac{p\bar{v}}{\bar{R}T} = \frac{pv}{RT}$$

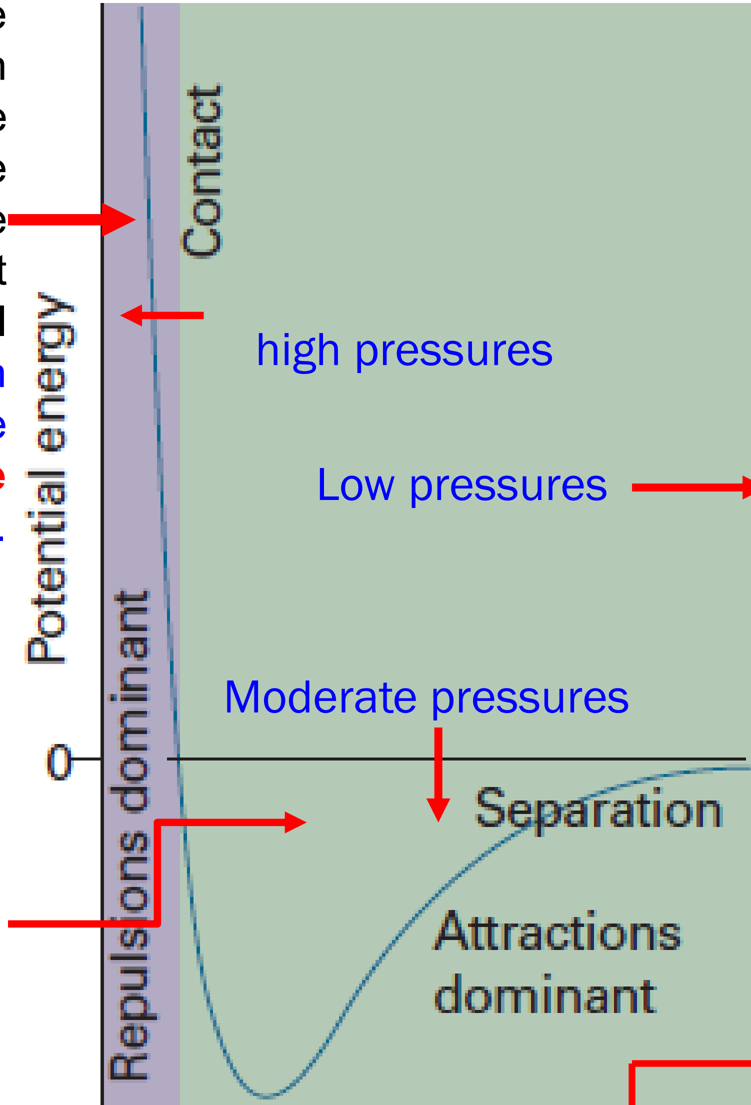


The compressibility factor Z tends to unity as pressure tends to zero at fixed temperature. This can be illustrated by reference to Fig. which shows Z for hydrogen plotted versus pressure at a number of different temperatures. In general, at states of a gas where pressure is small relative to the critical pressure, Z is approximately 1.

$$\lim_{p\to 0} Z = 1$$

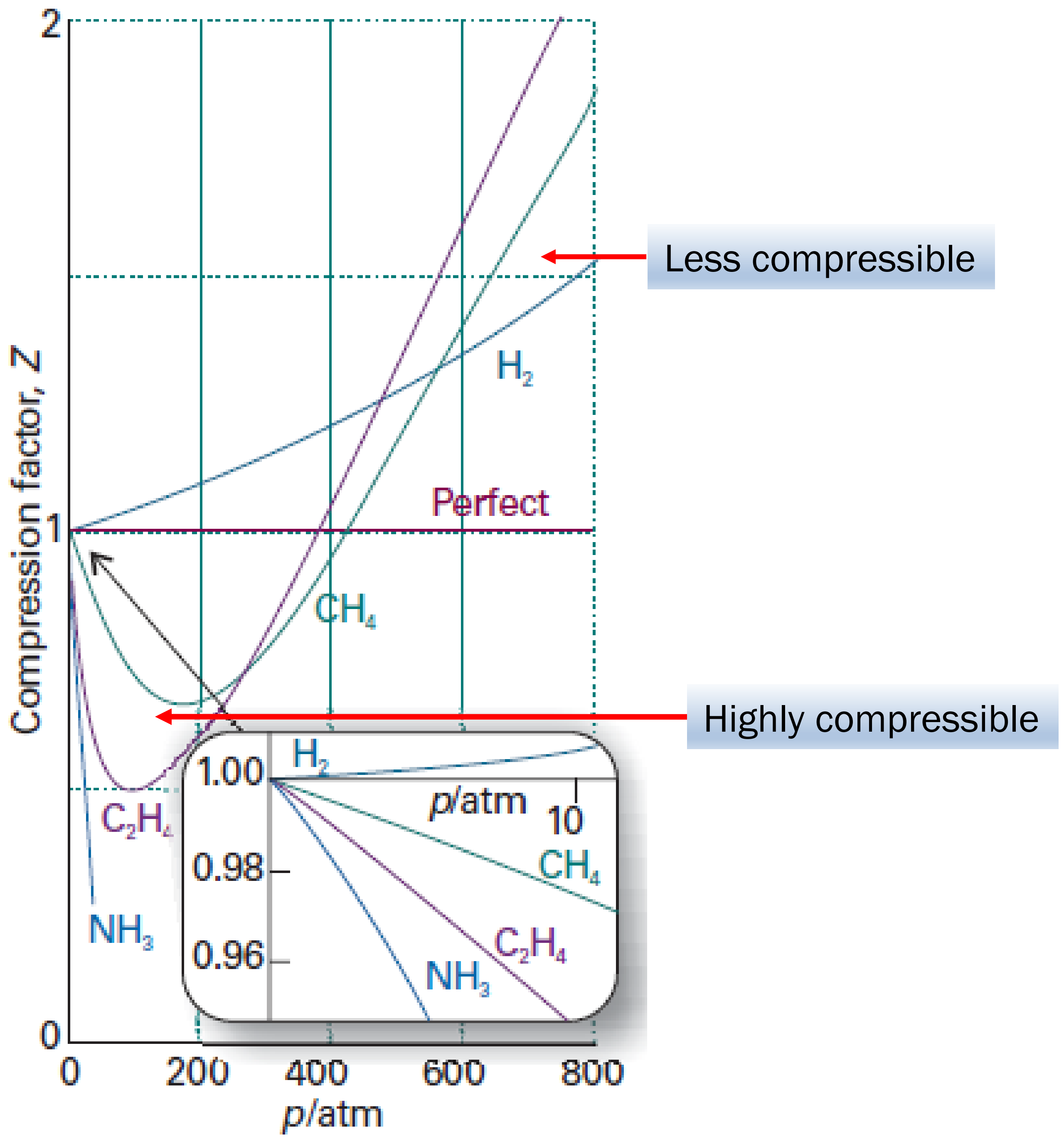


Repulsive forces are significant only when molecules are almost in contact: they are short-range interactions, even on a scale measured in molecular diameters Because they are short-range interactions, repulsions can be expected to be important only when the average separation of the molecules is small. This is the case at high pressure, when many molecules occupy a small volume. At high pressures, when the average separation of the molecules is small, the repulsive forces dominate and the gas can be expected to be **less compressible** because now the forces help to drive the molecules apart. $(Z > 1)$



On the other hand, attractive intermolecular forces have a relatively long range and are effective over several molecular diameters. They are important when the molecules are fairly close together but not necessarily touching (at the intermediate separations. At moderate pressures, when the average separation of the molecules is only a few molecular diameters, the attractive forces dominate the repulsive forces. In this case, the gas can be expected to be more compressible than a perfect gas because the forces help to draw the molecules together $(Z < 1)$

At low pressures, when the sample occupies a large volume, the molecules are so far apart for most of the time that the intermolecular forces play no significant role, and the gas behaves virtually perfectly. $(Z = 1)$



$$Pv = ZRT$$

$$\left(\frac{\partial v}{\partial P}\right)_T = -\frac{ZRT}{P^2}$$

Law of corresponding states

Figure 1 gives the compressibility factor for hydrogen versus pressure at specified values of temperature. Similar charts have been prepared for other gases. When these charts are studied, they are found to be qualitatively similar. Further study shows that when the coordinates are suitably modified, the curves for several different gases coincide closely when plotted together on the same coordinate axes, and so quantitative similarity also can be achieved. This is referred to as the principle of corresponding states. In one such approach, the compressibility factor Z is plotted versus a dimensionless reduced pressure P_r and reduced temperature T_r, defined as

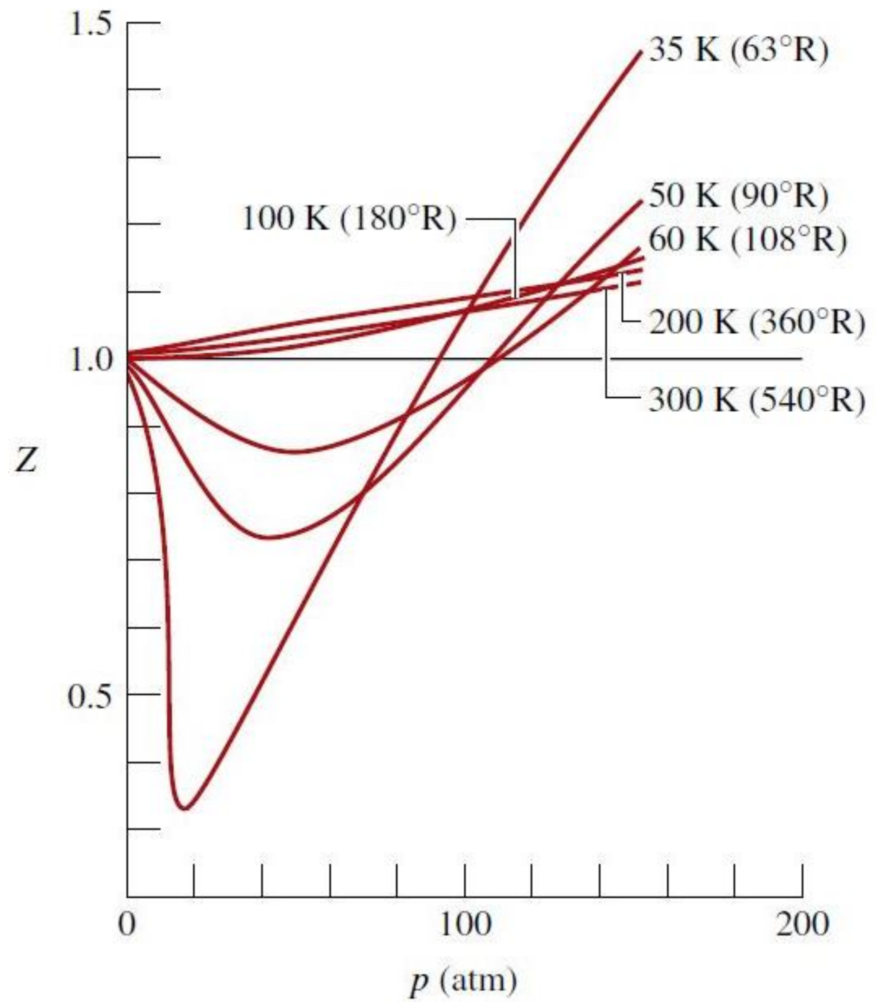
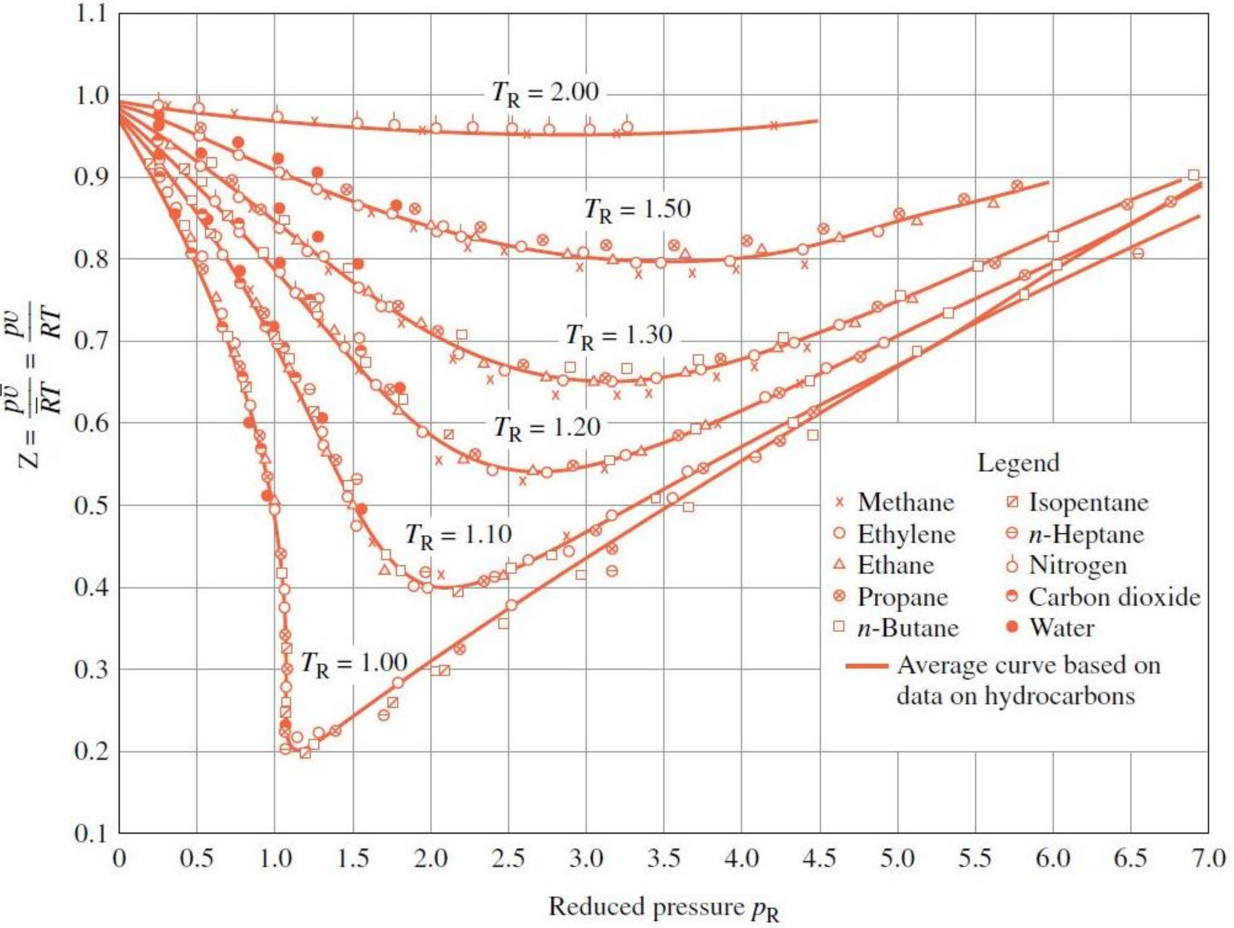


Figure 1

Law of Corresponding States

The law of corresponding states specifies that all fluids when compared at the same reduced temperature and reduced pressure, have approximately the same compressibility factor and all deviate from ideal gas behavior to about the same degree .



$$P_r = \frac{P}{P_{cr}}$$

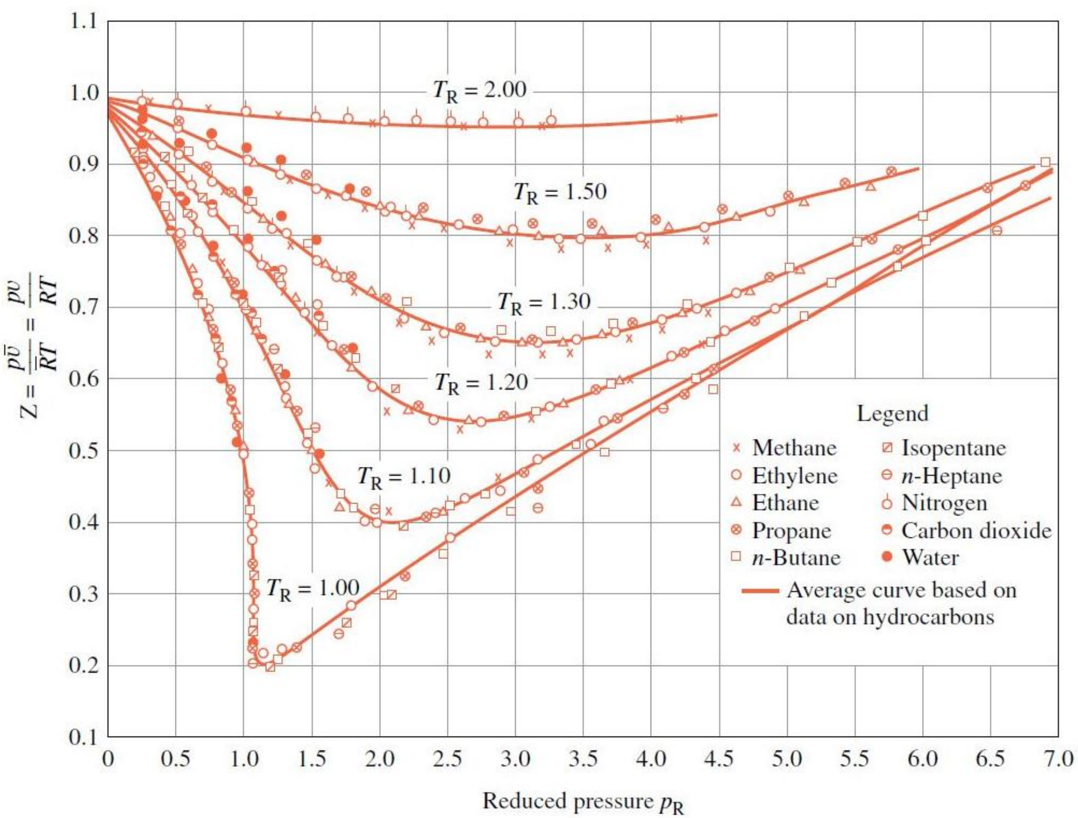
$$T_r = \frac{T}{T_{cr}}$$

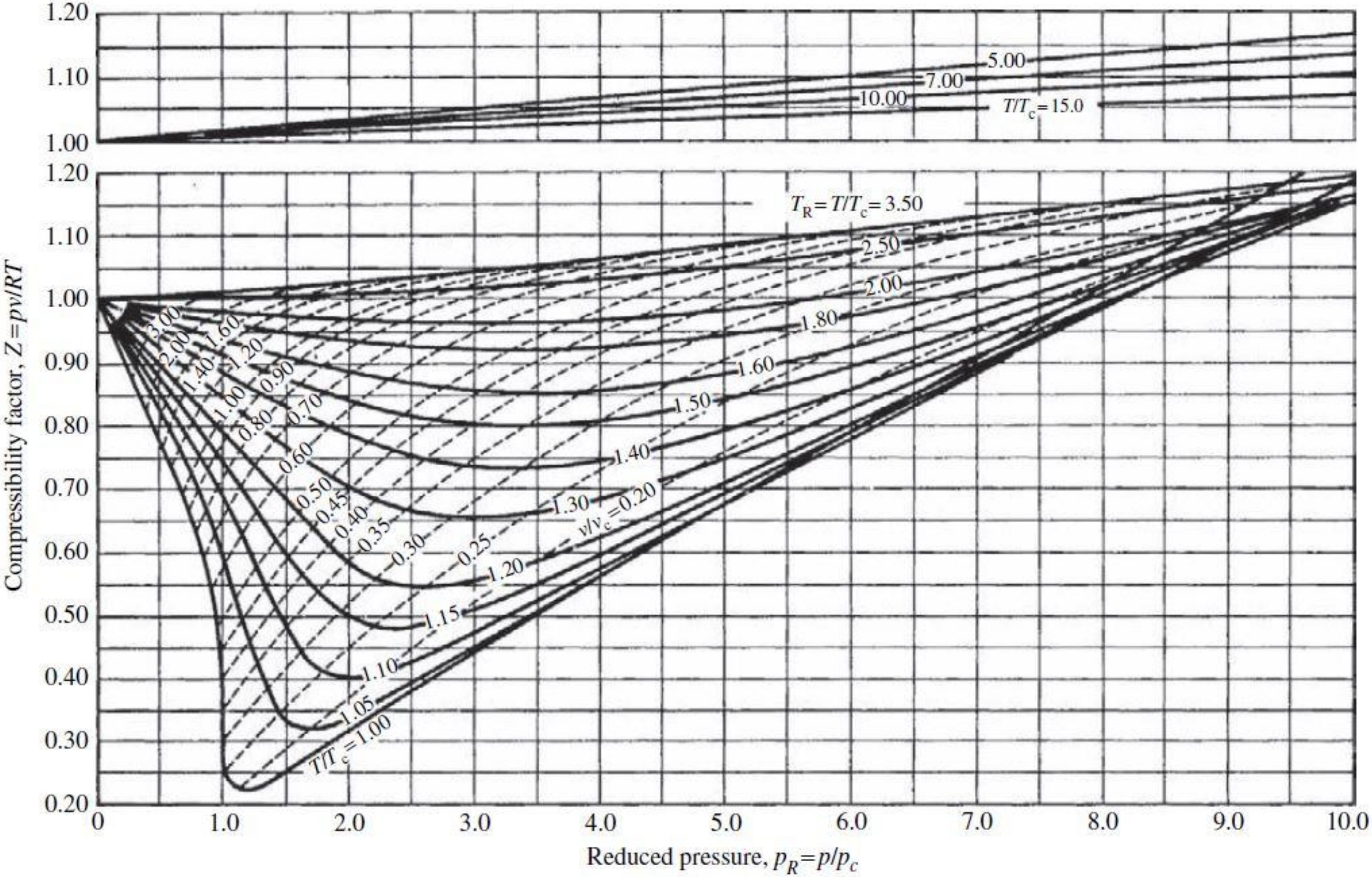
$$v_r = \frac{v}{v_{cr}}$$

Use of the generalized compressibility chart requires the knowledge of critical point data $\mathbf{0 < P_r < 1.0}$ $\mathbf{0 < P_r < 7.0}$ $\mathbf{0 < P_r < 40.0}$

From the chart it can be observed that:

- At very low pressure $P_r \ll 1, Z \approx 1.0, for\ all\ T_r$
- At high temperature $T_r > 2, Z \approx 1.0, for\ all\ P_r\ except P_r \gg 1.0$
- The deviation is greater near the critical points





In this figure a pseudo critical specific volume is used to calculate a pseudo reduced specific volume

$$v'_c = \frac{RT_c}{p_c} \qquad v'_r = \frac{v}{v'_c} = \frac{vp_c}{RT_c}$$

van der Waals equation of state:

$$\left(p + \frac{a}{\bar{v}^2}\right)(\bar{v} - b) = RT$$

where a and b are constants that account for the intermolecular forces and volume of the gas molecules respectively.

b is the volume per mole of the molecules if they were all crowded together, it is of the same order of magnitude of the molar specific volume of the liquid phase of the substance. Van der Waal reasoned that the pressure of real gas is less than that of an ideal gas at the same temperature and density because of the attractive forces between the molecules, and this reduction in pressure is proportional to the square of the density given by:

$$p = \frac{RT}{(\bar{v} - b)} - \frac{a}{\bar{v}^2}$$

Notice that the effect of each modification is greatest for state of high density

The critical point properties can be related to the van der Waals constants.

Evaluating a and b

At the critical point the two volumes (i.e. the v_g and v_f) coincide . On a p-v diagram the critical point is the limiting position as two points lying on a horizontal line approach each other. Hence, at the critical point, the critical isotherm has a horizontal tangent (point of inflection). This can be expressed mathematically as:

$$\left(\frac{\partial p}{\partial \bar{v}}\right)_{T=Tc} = 0; \quad \left(\frac{\partial^2 p}{\partial \bar{v}^2}\right)_{T=Tc} = 0$$

Applying the above two equations with the van der Waal equation

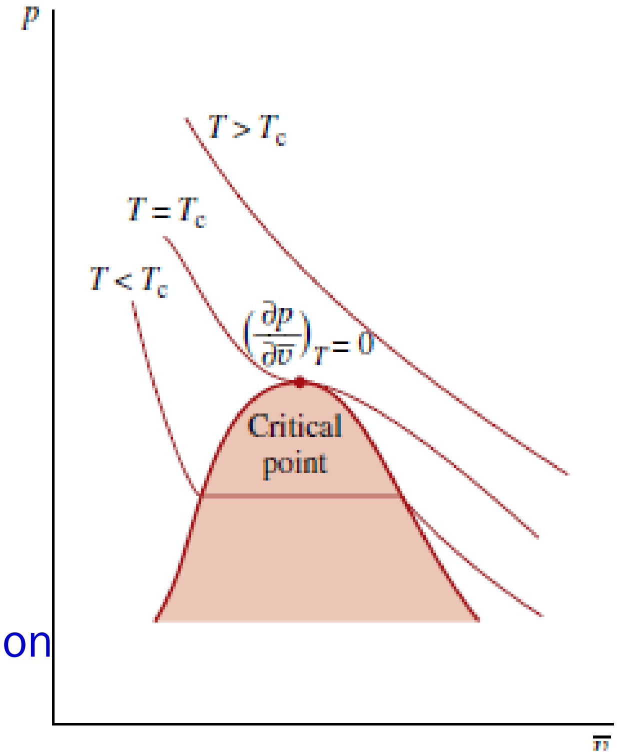
$$p_c = \frac{\bar{R}T_c}{(\bar{v}_c - b)} - \frac{a}{\bar{v}_c^2}$$

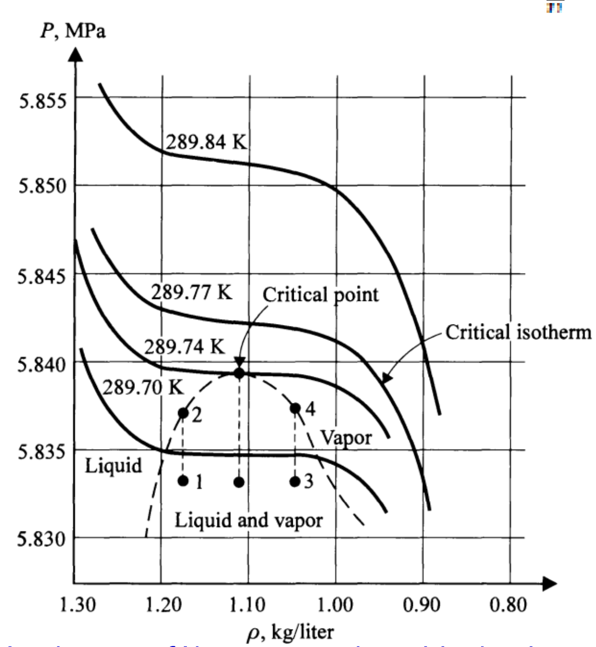
$$\left(\frac{\partial p}{\partial \bar{v}}\right)_{T=Tc} = -\frac{\bar{R}T_c}{(\bar{v}_c - b)^2} + \frac{2a}{\bar{v}_c^3} = 0 \ldots\ldots(2)$$

$$\left(\frac{\partial^2 p}{\partial \bar{v}^2}\right)_{T=Tc} = \frac{2\bar{R}T_c}{(\bar{v}_c - b)^3} - \frac{6a}{\bar{v}_c^4} = 0 \ldots\ldots(3)$$

Solving the above three equations it can be shown that:

$$\bar{v}_c = 3b \;\; ; \; p_c = \frac{a}{27b^2} \;\; ; \; T_c = \frac{8a}{27\bar{R}b}$$





Isotherms of Xenon near the critical point

Dieterici (1899) Equation of State of Real gas

$$p(v - b) = RT exp\left[-a/RTv\right]$$

Berthelot (1899) Equation of State of Real gas

$$p(v - b) = RT - \frac{a}{T}\left(\frac{v - b}{v^2}\right)$$

The most useful, best known, and most accurate equation of state for real gases are Beattie-Bridgeman (1928)

$$p = \left(\frac{1 - \epsilon}{v^2}\right)(v + B)RT - \frac{A}{v^2}$$

$$A = A_0(1 - a/v) \qquad B = B_0(1 - b/v) \qquad \varepsilon = \frac{c}{vT^3}$$

Redlich-Kwong equation:

$$p = \frac{\bar{R}T}{(\bar{v} - b)} - \frac{a}{\sqrt{T}\bar{v}(\bar{v} + b)}$$

The a and b in the Redlich-Kwong equation are not the same as those in the van der Waals equation. The two empirical constants can be obtained from critical-state data as:

$$a = 0.427 \frac{\bar{R}^2 T_c^{2.5}}{p_c} \qquad b = 0.0866 \frac{\bar{R}T_c}{p_c}$$

A virial equation is more generalized form of equation of state. It is written as:

$$p\bar{v} = \bar{R}T\left(1 + \frac{B}{\bar{v}} + \frac{C}{\bar{v}^2} + \cdots\right)$$

where B,C,... are all empirically determined functions of temperature and are called as virial coefficients. The word virial stems from the **Latin word for force.** In the present usage it is force interactions among molecules that are intended.

Virial Coefficients

At large molar volumes and high temperatures, the real-gas isotherms do not differ greatly from perfect-gas isotherms. The small differences suggest that the perfect gas law is in fact the first term in an expression of the form

$$p\bar{v} = \bar{R}T(1 + B'p + C'p^2 + \cdots)\text{....(a)}$$

A more convenient expansion for many applications is

$$p\bar{v} = \bar{R}T\left(1 + \frac{B}{\bar{v}} + \frac{C}{\bar{v}^2} + \cdots\right)\text{......(b)}$$

By comparing the equations a and b with the equation c we see that the term in parentheses can be identified with the compression factor, *Z*.

$$p\bar{v} = Z\bar{R}T \;\ldots..(c)$$

The coefficients *B*, *C*, . . . , which depend on the temperature, are the second, third, . . . **virial coefficients**; the first virial coefficient is 1. The third virial coefficient, *C*, is usually less important than the second coefficient, *B*, in the sense that at typical molar volumes $\frac{C}{\bar{v}^2} \ll \frac{B}{\bar{v}}$

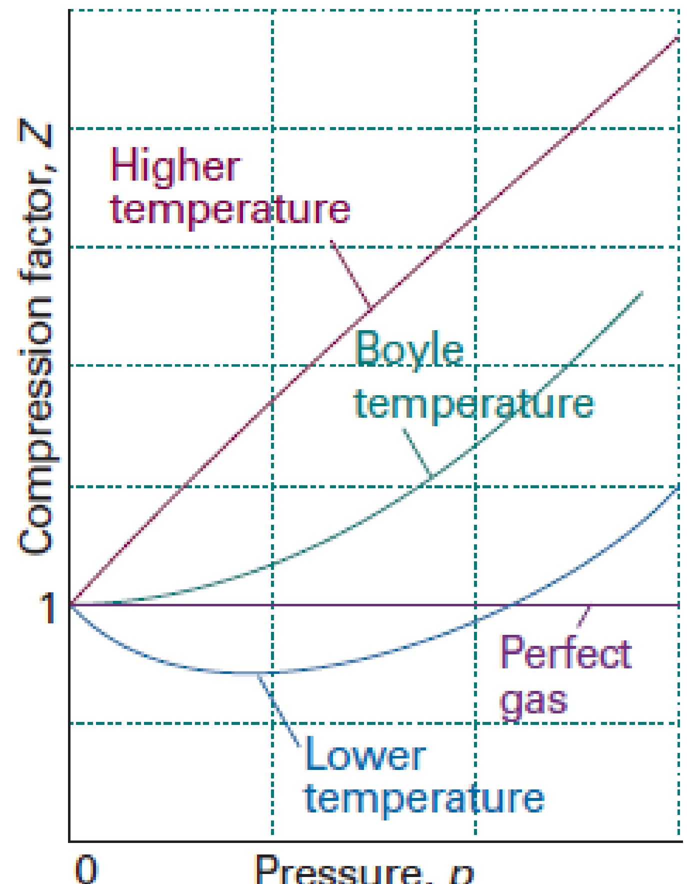
We can use the virial equation to demonstrate the important point that, although the equation of state of a real gas may coincide with the perfect gas law as $p \to 0$, not all its properties necessarily coincide with those of a perfect gas in that limit. Consider, for example, the value of dZ/dp, the slope of the graph of compression factor against pressure. For a perfect gas $dZ/dp = 0$ (because $Z = 1$ at all pressures), but for a real gas from eqn. a we obtain

$$p\bar{v} = \bar{R}T(1 + B'p + C'p^2 + \cdots)....(a)$$

$$Z = (1 + B'p + C'p^2 + \cdots) \qquad \frac{dZ}{dp} = B' + 2C'p + \cdots$$

$$p\bar{v} = \bar{R}T\left(1 + \frac{B}{\bar{v}} + \frac{C}{\bar{v}^2} + \cdots\right)......(b)$$

$$Z = \left(1 + \frac{B}{\bar{v}} + \frac{C}{\bar{v}^2} + \cdots\right) \qquad \frac{dZ}{d(1/\bar{v})} = B + \frac{2C}{\bar{v}} + \cdots$$



- At the **Boyle temperature**, T_B, the properties of the real gas do coincide with those of a perfect gas as $p \to 0$.
- **$B = 0$ at the Boyle temperature.**
- $p\bar{v} = \bar{R}T_B$ over a more extended range of pressures than at other temperatures because the first term after 1 (that is, $\frac{B}{\bar{v}}$ in the virial equation is zero and $\frac{C}{\bar{v}^2}$ and higher terms are negligibly small.
- For helium T_B = 22.64 K; for air T_B = 346.8 K